

RADIUS VARIATION AND GENERALIZATION IN PRE-NORMALIZED TRANSFORMERS

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ABSTRACT

Pointwise Lipschitz continuity does not by itself justify removing normalization from a vector Rademacher bound. For the process with one sign per example, we show that RMS normalization on an annulus of half-width ρ has worst-case distortion $\Theta(1 + \rho\sqrt{m})$, for fixed regularizer and bounded ρ . A profile inequality recovers constant-one contraction when each example’s radius is fixed across hypotheses. We connect these evaluation-space results to pre-normalized transformers through an explicit parameter-to-radius recursion and a local sequence-loss covering bound around an independent reference. We then audit the construction and the certificates. Across seven trained decoder configurations and three training/data seeds, all 612 finite-family stage diagnostics remain below the sampled-floor amplification threshold; every pointwise sign-bootstrap upper endpoint also remains below it. In contrast, no tested positive-floor candidate is certified, and all 60 evaluated generalization bounds are vacuous. The mean logarithm of the zero-floor propagation product is nearly linear in depth, while residual scaling and a larger regularizer reduce its severity without yielding a useful certificate. These findings distinguish a sharp worst-case obstruction from the conservativeness of this explicit propagation argument. The lower bound concerns a specified vector process and decoder subclass; the empirical diagnostics concern finite random families, not a uniform contraction theorem for trained models.

1 INTRODUCTION

Normalization appears at every sublayer of a pre-normalized transformer, but its role in capacity analysis is easy to misidentify. A small Lipschitz constant controls the distance between two normalized vectors. A Rademacher supremum asks a different question: after observing random signs, which hypothesis makes their signed sum largest? Because a hypothesis determines both the direction and the radius of each hidden state, normalization can alter that choice. A radius floor controls the first question without resolving the second.

We isolate this issue for the Euclidean norm of a signed vector sum, with one sign per example. On a common annulus of half-width ρ , we prove that its worst-case normalization cost is of order $1 + \rho\sqrt{m}$. Thus an arbitrarily thin fixed annulus does not imply a contraction uniform in sample size and dimension. The threshold for such a worst-case guarantee is $\rho = O(m^{-1/2})$. A construction with a shared direction and independently variable orthogonal components gives the lower bound. It can be realized exactly by a restricted learned-embedding decoder class.

For transformers, an annulus assumption alone leaves a practical gap: what parameter constraints imply the required control of hidden-state radii? We provide a quantitative answer for neighborhoods of a reference model. Layerwise perturbation budgets generate an explicit error recursion. Its solution controls variation around the reference radius profile and can certify a population radius floor for every model in the neighborhood. The same parameter budgets enter a covering bound for the complete decoder. This connects a normalization theorem about evaluation classes to a uniform generalization theorem about a specified parameter class. Whether the resulting sufficient certificate is informative at trained references is a separate question, which we evaluate rather than assume.

The contributions are: (i) a sharp worst-case annulus theorem and an additive radius-profile bound (Section 3); (ii) an explicit, auditable parameter-to-radius certificate and local entropy/generalization

bound (Sections 4–5); and (iii) a controlled empirical study separating random-neighborhood behavior from certificate looseness (Section 6). Seven configurations, covering depth, width, residual scaling, and the RMS regularizer, are evaluated under three independently seeded training/data runs. Random local families show no sampled-floor-normalized amplification, but all tested positive-floor certificates fail and all evaluated loss bounds remain vacuous. The nearly linear growth of $\log_{10} \Pi_0$ with depth identifies accumulated uniform propagation as a major source of conservativeness in this bound. Complete proofs, architectural constants, and experimental details are in the appendix.

The two conclusions have different logical status. The sharp theorem proves that a particular dimension-uniform contraction strategy can fail. The experiments establish a repeatable finite-family observation on a controlled task, together with a numerical limitation of our sufficient certificate. They do not establish a generalization lower bound or exclude sharper certificates for the same trained models.

Relation to prior work. RMSNorm and LayerNorm divide by different radial quantities (Zhang & Sennrich, 2019; Ba et al., 2016). Our LayerNorm implication is restricted to the mean-zero subspace. The process studied here is different from the coordinate-indexed process in vector contraction (Maurer, 2016); our counterexample does not contradict that inequality. Linear covers and optimized layerwise entropy allocation are established tools (Zhang, 2002; Bartlett et al., 2017; Edelman et al., 2022). We use them with explicit normalization and perturbation certificates, rather than claim these tools as new.

Trauger & Tewari (2024) remove explicit sequence-length dependence under their norm assumptions. Our Euclidean-input cover retains $\log(NT)$ and does not improve that dependence. Li et al. (2026) study excess risk using offset complexity, while Sakai & Imaizumi (2026) obtain spectrum-adaptive guarantees with post hoc selection of Schatten indices. Their normalized block definitions differ from the pre-RMS residual decoder studied here, and their objectives and admissible class selections differ from our fixed-reference result. We claim neither an improved general rate nor a flaw in those bounds. Our focus is the sharp normalization obstruction and a constructive route from weight perturbations to radius control and loss entropy.

2 SETUP AND THE CONTRACTION QUESTION

For row vectors $h \in \mathbb{R}^d$ and streams $H \in \mathbb{R}^{T \times d}$, write $\|h\|_{\circ} = \|h\|_2 / \sqrt{d}$ and $\|H\|_{\infty, \circ} = \max_t \|H_t\|_{\circ}$. The mixed norm is $\|W\|_{2,1} = \sum_j \|W_{\cdot j}\|_2$; it sums *columns* in our row-vector convention. Fix $\nu > 0$ and define

$$\mathcal{N}(h) = \frac{h}{\sqrt{\|h\|_{\circ}^2 + \nu}}, \quad \text{RMS}_{\gamma}(h) = \mathcal{N}(h) \text{diag}(\gamma). \quad (1)$$

For a nonempty class $\mathcal{H} = \{(h_i(\theta))_{i=1}^m : \theta \in \Theta\}$, let

$$\mathfrak{R}_m^{\text{vec}}(\mathcal{H}) = \frac{1}{m} \mathbb{E}_{\varepsilon} \sup_{\theta \in \Theta} \left\| \sum_{i=1}^m \varepsilon_i h_i(\theta) \right\|_2. \quad (2)$$

There is one independent sign per evaluation. This complexity can be positive for a singleton class. Scalar empirical Rademacher complexity, denoted $\widehat{\mathfrak{R}}_m$, uses $m^{-1} \mathbb{E}_{\varepsilon} \sup_f \sum_i \varepsilon_i f_i$, without an absolute value. All classes are measurable and separable; bounded finite-dimensional parameter sets can be replaced by their closures.

If all radii are at least $c \geq 0$, the gain-free normalizer is $b_c = (c^2 + \nu)^{-1/2}$ -Lipschitz on that region. This follows from Lemma 4.1 below. The tempting inference is

$$\mathfrak{R}_m^{\text{vec}}(\mathcal{N} \circ \mathcal{H}) \leq C b_c \mathfrak{R}_m^{\text{vec}}(\mathcal{H}) \quad \text{with a constant } C \text{ independent of } m, d, \mathcal{H}. \quad (3)$$

Section 3 characterizes precisely how this inference fails on common annuli. The result concerns a proposed proof step, not generalization of a trained decoder’s loss.

3 THE SHARP COST OF RADIUS VARIATION

Theorem 3.1 (Contraction with a reference radius profile). *Suppose $\|h_i(\theta)\|_0 \geq c$, $\|h_i(\theta)\|_2 \leq M$, and $\|\gamma(\theta)\|_\infty \leq G$. If numbers r_i^0 , fixed across θ , satisfy $|\|h_i(\theta)\|_0 - r_i^0| \leq \rho$, then*

$$\mathfrak{R}_m^{\text{vec}}(\text{RMS}_\gamma \circ \mathcal{H}) \leq G\{b_c \mathfrak{R}_m^{\text{vec}}(\mathcal{H}) + M\rho b_c^2\}. \quad (4)$$

In particular, fixed radii across hypotheses give the constant-one bound $Gb_c \mathfrak{R}_m^{\text{vec}}(\mathcal{H})$.

Proof idea. Clip the reference radii from below at c and write $q_i(\theta) = (\|h_i(\theta)\|_0^2 + \nu)^{-1/2} = q_i^0 + \Delta q_i(\theta)$. Then $q_i^0 \leq b_c$ and $|\Delta q_i| \leq \rho b_c^2$. Fixed-coordinate contraction controls the $q_i^0 h_i$ term; the triangle inequality controls the remaining signed sum by $mM\rho b_c^2$. The gain is a common operator for each hypothesis. See Appendix A for details. $\square$

The profile may vary arbitrarily between examples; the condition controls variation *between hypotheses at the same example*. Measuring the radii of one checkpoint across tokens does not measure this quantity.

For $\rho \geq 0$, define

$$C_{m,\nu}^*(\rho) = \sup_{\substack{d \geq 2, \mathcal{H} \neq \emptyset \\ 1 \leq \|h_i(\theta)\|_0 \leq 1+2\rho \ \forall i, \theta}} \frac{\sqrt{1+\nu} \mathfrak{R}_m^{\text{vec}}(\mathcal{N} \circ \mathcal{H})}{\mathfrak{R}_m^{\text{vec}}(\mathcal{H})}. \quad (5)$$

The denominator is positive under the stated floor.

Theorem 3.2 (Sharp annulus scaling). *Fix $\nu > 0$ and $\rho_0 > 0$. There are explicit positive constants c_{ν,ρ_0} and C_{ν,ρ_0} such that, for every $m \geq 1$ and $0 \leq \rho \leq \rho_0$,*

$$\max\{1, c_{\nu,\rho_0} \rho \sqrt{m}\} \leq C_{m,\nu}^*(\rho) \leq 1 + C_{\nu,\rho_0} \rho \sqrt{m}. \quad (6)$$

Consequently $C_{m,\nu}^(\rho) = \Theta_{\nu,\rho_0}(1 + \rho \sqrt{m})$. For bounded sequences ρ_m , the worst-case distortion is uniformly bounded exactly when $\sup_m \rho_m \sqrt{m} < \infty$.*

Proof idea and the lower construction. For every admissible class, the Hilbert-space moment bound in Lemma A.1 gives $\mathfrak{R}_m^{\text{vec}}(\mathcal{H}) \geq \sqrt{d/(3m)}$. Apply Theorem 3.1 with $r_i^0 = 1 + \rho$ and $M = \sqrt{d}(1 + 2\rho)$ to obtain the upper bound. For the lower bound, set $d = m + 1$ and choose an orthonormal basis $e_0, \dots, e_m$. With $t^2 = 4\rho(1 + \rho)$, let

$$h_i(a) = \sqrt{d}(e_0 + a_i e_i), \quad a \in \{0, t\}^m. \quad (7)$$

Every radius is 1 or $1 + 2\rho$. If $S = \sum_i \varepsilon_i$, then

$$\mathfrak{R}_m^{\text{vec}}(\mathcal{H}_{m,t}) = \frac{\sqrt{d}}{m} \mathbb{E} \sqrt{S^2 + t^2 m} \leq \sqrt{\frac{d(1+t^2)}{m}}. \quad (8)$$

Choose $a_i = 0$ for positive signs and $a_i = t$ for negative signs. The expected projection of the normalized signed sum onto e_0 , divided by m , is

$$\frac{\sqrt{d}}{2} \left\{ (1 + \nu)^{-1/2} - (1 + t^2 + \nu)^{-1/2} \right\}. \quad (9)$$

The norm dominates this projection. Its ratio to (8), with the normalization in (5), is at least $c_{\nu,\rho_0} \rho \sqrt{m}$. A singleton radius-one class gives the additional lower bound one. Explicit constants and all endpoint cases are proved in Appendix A. $\square$

The construction illustrates the obstruction: before normalization the shared component is independent of a , whereas after normalization its coefficient can respond to the signs. The orthogonal components keep the unnormalized supremum small enough for the ratio to grow.

Realization and scope. Use the vectors in (7) as the independently selectable rows of a learned embedding table, take length-one inputs consisting of m distinct tokens, and set every residual branch to zero. This realizes the class exactly at every depth, with embedding RMS row budget $1 + 2\rho$ and a population radius floor of one on those tokens (Proposition A.2). Here $d = m + 1$ and the vocabulary grows with m . This is not a fixed-width lower bound. It also does not transfer to the entire norm-ball decoder class: a ratio of two complexities need not increase under class enlargement. The same construction in the mean-zero subspace of $\mathbb{R}^{m+2}$ applies to LayerNorm. None of these statements is a lower bound for the bounded loss used below.

4 FROM WEIGHT PERTURBATIONS TO RADIUS PROFILES

4.1 DECODER AND LOCAL PARAMETER CLASS

Consider a decoder with learned embeddings $H_t^0 = E_{X_t, \cdot}$ and $D = 2L$ residual sublayers,

$$H^s = H^{s-1} + \lambda_s F_s(\mathcal{N}(H^{s-1}); W_s), \quad \lambda_s \geq 0 \text{ fixed.} \quad (10)$$

Branches alternate masked multi-head attention and $F_{\text{ff}}(Z) = [\text{swish}(Z A_g) \odot Z A_u] W_d$, where $\text{swish}(x) = x/(1 + e^{-x})$ (Shazeer, 2020). Attention uses separate query, key, value, and output matrices with the usual $1/\sqrt{d_h}$ score scaling. The main model has ordinary multi-head attention; grouped queries and fixed orthogonal rotary maps are included in Appendix C.2. There are no additive biases, dropout, or learned position maps. Normalization gains are absorbed into the first branch matrices, exactly once. Norm bounds below concern these effective matrices.

On inputs with $\|Z\|_{\infty, \circ} \leq 1$, fixed ambient matrix caps give a branch Lipschitz constant Λ_s . For each linear stage j in branch $s(j)$, b_j bounds its input Euclidean row norm and ℓ_j converts a linear output error into branch RMS error. Appendix Table 2 gives all seven certificates for attention and SwiGLU, with proofs. In particular, changing only stage j by spectral distance τ_j changes the branch output by at most $\ell_j b_j \tau_j$. The caps hold for every hybrid choice of matrices.

Fix a reference parameter $\theta^0 = (E^0, (W_j^0)_j, C^0)$ inside the ambient caps. Its neighborhood is defined by

$$\begin{aligned} \max_v \|E_v - E_v^0\|_{\circ} &\leq e_E, & \|W_j - W_j^0\|_{\text{op}} &\leq \tau_j, & \|W_j - W_j^0\|_{2,1} &\leq R_j^{\text{loc}}, \\ \max_v \|C_v - C_v^0\|_2 &\leq e_C. \end{aligned} \quad (11)$$

Intersect these conditions with the ambient caps, including $\max_v \|E_v\|_{\circ} \leq B_0$ and $\max_v \|C_v\|_2 \leq r_{\text{out}}$. All reference choices and budgets are fixed independently of the N sequences used for the generalization guarantee. A reference trained on independent pilot data is allowed by conditioning on those data. Parameter couplings, including tied readouts, can be retained by restricting this class; covers use its independent-matrix relaxation.

4.2 A CERTIFICATE THAT DOES NOT ASSUME ITS OWN CONCLUSION

Lemma 4.1 (Global extension of normalization). *For $c \geq 0$, the map*

$$\overline{\mathcal{N}}_c(h) = \frac{h}{\sqrt{\max\{\|h\|_{\circ}^2, c^2\} + \nu}} \quad (12)$$

agrees with $\mathcal{N}$ above radius c , has output RMS radius at most one, and is globally $(c^2 + \nu)^{-1/2}$ -Lipschitz.

The extension is a proof device. It lets us propagate a candidate floor through approximate or perturbed models before establishing that those models satisfy the floor. Its radial proof is in Appendix B.

Theorem 4.2 (Parameter-to-radius certificate). *Suppose the reference satisfies $\|H_t^s(X; \theta^0)\|_{\circ} \geq a_s$ for every input in a specified set $\mathcal{X}$, every position, and $s = 0, \dots, D$. Choose $0 \leq c_s \leq a_s$ and define*

$$\omega_s = (c_s^2 + \nu)^{-1/2}, \quad L_s = 1 + \lambda_s \omega_{s-1} \Lambda_s, \quad q_s = \sum_{j:s(j)=s} \ell_j b_j \tau_j. \quad (13)$$

Let $\Delta_0 = e_E$ and

$$\Delta_s = L_s \Delta_{s-1} + \lambda_s q_s. \quad (14)$$

If $\Delta_s \leq a_s - c_s$ for every s , then every model in the neighborhood satisfies, uniformly on $\mathcal{X}$,

$$\|H^s(X; \theta) - H^s(X; \theta^0)\|_{\infty, o} \leq \Delta_s, \quad (15)$$

$$|\|H_t^s(X; \theta)\|_o - \|H_t^s(X; \theta^0)\|_o| \leq \Delta_s, \quad \|H_t^s(X; \theta)\|_o \geq c_s. \quad (16)$$

Alternatively, compute (14) with every $c_s = 0$. Then the stream and profile estimates always hold, and $\max\{a_s - \Delta_s, 0\}$ is a valid floor without a feasibility test.

Proof idea. Replace every normalizer by its extension for the proposed floor. Compare the extended candidate and the reference first at identical weights, then telescope the weight changes at the fixed normalized reference input. The two errors are $L_s \Delta_{s-1}$ and $\lambda_s q_s$. Reverse triangle gives radius at least $a_s - \Delta_s \geq c_s$. The extended and original recursions therefore coincide, by induction. For zero floors the extension is already the original normalizer. Appendix B supplies the full argument. $\square$

This is a population certificate only when $\mathcal{X}$ contains the population support and the reference floors hold there. On a finite test set, it is a deterministic certificate on that set. One useful reference with an explicit population floor has nonzero embedding rows and zero branches; Appendix B gives a concrete nontrivial neighborhood around it. No empirical minimum is promoted to a population floor.

Corollary 4.3 (A sufficient local scale for contraction). *Let $B_s = B_0 + \sum_{u=1}^s \lambda_u \Lambda_u$ and suppose Theorem 4.2 certifies $c_s > 0$. On any m evaluations of the stage- s hidden-state class $\mathcal{H}_s$,*

$$\mathfrak{R}_m^{\text{vec}}(\text{RMS}_\gamma \circ \mathcal{H}_s) \leq G \omega_s \left(1 + \frac{\sqrt{3} B_s \omega_s \Delta_s \sqrt{m}}{c_s} \right) \mathfrak{R}_m^{\text{vec}}(\mathcal{H}_s). \quad (17)$$

If $e_E = \eta e$, $\tau_j = \eta t_j$, and the ambient caps and certified floors stay fixed, then $\Delta_s = \eta D_s$ for a fixed explicit recursion D_s . For fixed s and these constants, $\eta = O(m^{-1/2})$ is sufficient for bounded contraction.

This is a sufficient condition on parameter neighborhoods, not a necessity theorem for them. The necessity in Theorem 3.2 concerns the larger family of all evaluation classes in a common annulus. In Section 6, the positive-floor sufficient condition fails throughout the tested neighborhoods; the unconditional zero-floor recursion remains available and its slack is explicitly reported.

5 GENERALIZATION BY COVERING LOCAL PARAMETER CLASSES

Let the final logits be $z_t = \mathcal{N}(H_t^D)C$. Apply the fixed coordinatewise cap $\psi(z) = c_{\log} \tanh(z/c_{\log})$ and use the sequence-averaged cross-entropy

$$g_\theta(X, Y) = \frac{1}{T} \sum_{t=1}^T \left[\log \sum_{v=1}^V e^{\psi(z_{t,v})} - \psi(z_{t,Y_t}) \right]. \quad (18)$$

It lies in $[0, B]$ for $B = \log(1 + (V-1)e^{2c_{\log}})$ and is 2-Lipschitz in the maximum logit error. There are N independent identically distributed sequences of length T ; positions within a sequence may be dependent. Use floors c_s certified by Theorem 4.2, or imposed as genuine population restrictions on the local class. Assume this class is nonempty. Zero floors are always admissible.

Write $\Pi_s = \prod_{u=s+1}^D L_u$ and define

$$\begin{aligned} \Pi_{\text{out}} &= 2r_{\text{out}} \sqrt{d} \omega_D, & A_{\text{loc}} &= 4e_E \Pi_{\text{out}} \Pi_0, \\ w_j &= \Pi_{\text{out}} \Pi_{s(j)} \lambda_{s(j)} \ell_j b_j R_j^{\text{loc}}, & w_C &= 2\sqrt{V} e_C \sqrt{d}, \\ K_{\text{loc}} &= \left(\sum_j w_j^{2/3} + w_C^{2/3} \right)^{3/2}. \end{aligned} \quad (19)$$

The total readout cap r_{out} controls propagation, whereas its perturbation budget e_C controls readout entropy. Likewise Λ_s, b_j, ℓ_j come from ambient caps, but R_j^{loc} controls stage entropy. Let V_{eff} count distinct input tokens in the sample, $P = \sqrt{V_{\text{eff}}d}$, and $\mathcal{L} = \log(2k_*) + \log(2NT + 1)$, where $k_* \geq 2$ bounds every linear output width and V .

Theorem 5.1 (Local loss entropy and generalization). *For the fixed local class above, at every $\epsilon > 0$,*

$$\log \mathcal{N}_{\infty}(\mathcal{G}_{\text{loc}}, \epsilon) \leq P^2 \log(1 + A_{\text{loc}}/\epsilon) + \frac{1024K_{\text{loc}}^2 \mathcal{L}}{\epsilon^2}. \quad (20)$$

Here the metric is maximum absolute loss error on the N sequences and centers may be external. Set

$$\Phi_N = \min \left\{ B/2, \inf_{0 < \alpha \leq B} \left[4\alpha + \frac{12}{\sqrt{N}} \left\{ BP \sqrt{\log(1 + A_{\text{loc}}/\alpha)} + 32K_{\text{loc}} \sqrt{\mathcal{L} \log(B/\alpha)} \right\} \right] \right\}. \quad (21)$$

With probability at least $1 - \delta$, simultaneously for all $\theta \in \Theta_{\text{loc}}$,

$$\mathbb{E} g_{\theta} \leq \frac{1}{N} \sum_{i=1}^N g_{\theta}(X_i, Y_i) + \min \left\{ B, 2\Phi_N + 3B \sqrt{\frac{\log(2/\delta)}{2N}} \right\}. \quad (22)$$

Fixed embedding or readout parameters have zero corresponding entropy cost. If $r_{\text{out}} = 0$, the loss class is constant and $A_{\text{loc}} = K_{\text{loc}} = 0$ may be used.

Proof structure. Translate each matrix class by its reference and cover its perturbations on the current fixed design. A proper mixed-norm linear cover has log-cardinality at most $256(R_j^{\text{loc}} b_j / \eta_j)^2 \mathcal{L}$. Properness preserves every ambient cap. Conditional covers of successive layers form a finite tree. If E_s is the stream approximation error, then $E_s \leq L_s E_{s-1} + \lambda_s \sum_{j:s(j)=s} \ell_j \eta_j$. The extension makes this recurrence valid at every accuracy, even for centers whose original-network radii would violate the floors. Optimizing the allocation of the η_j gives the $2/3$ -power sum in (19). The embedding cover gives the first entropy term. Truncated entropy integration, symmetrization, and bounded differences on independent sequences give (22). Appendix G proves each step, including the translation and statistical conditioning. $\square$

Setting the reference matrices to zero, $e_E = B_0$, $e_C = r_{\text{out}}$, and $R_j^{\text{loc}} = R_j$ recovers the global norm-class cover in Appendix F; the spectral perturbation caps can be taken redundant. Positive floors for that global class must be separately imposed, since a zero reference supplies none. Taking $\alpha = B/\sqrt{N}$ yields $N^{-1/2}$ times logarithmic factors for fixed architecture and budgets. No replacement of N by NT is justified by sequence averaging.

What localization changes. If the embedding, stage mixed-norm, and readout perturbation budgets are multiplied by η while the ambient caps and floors remain fixed, then A_{loc} and K_{loc} are multiplied by η . The radius certificate also scales linearly when the spectral perturbation budgets do. Nevertheless, small profile variation alone is not a generalization guarantee: (20) depends on parameter budgets and absolute floors, not just the annulus width ρ .

Class selection and depth. An arbitrary reference chosen from these same N sequences is outside the fixed-reference theorem. A countable family of predetermined references and budgets is allowed with failure probabilities $\delta \pi_a$, $\sum_a \pi_a \leq 1$, as proved in Corollary F.3. Depth appears through both Π_s and the number of variable stages. A certified linear floor makes propagation polynomial, but does not remove stage accumulation (Appendix H). Such a floor is an additional hypothesis, not a conclusion of residual architecture or of the experiments below.

6 EXPERIMENTS: NORMALIZATION BEHAVIOR AND CERTIFICATE LOOSENESS

We evaluate the exact lower construction (E1), normalization of finite families near trained references (E2), the radius recursion (E3), and the complete loss bound (E4). The experiments report both successful calculations and failed sufficient certificates; they are not a demonstration of a nonvacuous transformer generalization guarantee.

Design and uncertainty. The controlled task has vocabulary eight and context length 32. After two uniform tokens, the next token equals $(x_{t-1} + 2x_{t-2}) \bmod 8$ with probability 0.9 and is uniform otherwise. For each seed, 12,000 independent pilot sequences train the reference for 12 epochs; independent fit and test sets contain 512 and 2,000 sequences. The decoder is bias-free pre-RMS attention with SwiGLU and untied embeddings/readout. The baseline has $L = 4, d = 128, d_h = 32, d_{ff} = 4d, \lambda = 1, \nu = 10^{-5}$. We vary $L \in \{2, 8\}, d \in \{64, 256\}$, scaled $L = 8$ with $\lambda = (2L)^{-1/2} = 1/4$, and $\nu = 10^{-3}$, giving seven configurations. The three seed groups share sequences and sign draws across architectures within each seed, but regenerate data, initialization, directions, and signs between seeds. The first seed is retained from an initial ten-configuration screen; the other two are new runs.

The architecture study uses 32 presampled paired directions, the reference, and scales $\eta \in \{10^{-3}, 10^{-2}\}$: 65 models per finite family. The direction bank is fixed before evaluation signs are drawn, and normalization gains are held fixed. E2 uses one position per sequence, 1,024 common Rademacher draws, and 400 paired sign-bootstrap resamples. The baseline additionally uses nested prefixes $m = 64, 128, 256, 512$. Unless stated otherwise, results below use $m = 512, \eta = 10^{-2}$ and report mean $\pm$ sample standard deviation across the three seeds. These are descriptive seed summaries, distinct from conditional, pointwise sign-bootstrap intervals. The 612 stage rows are not independent experimental replicates. Appendix I gives implementation and provenance details.

E1: deterministic sharp-scaling verification. An earlier exact 80-point binomial sweep agrees with exhaustive enumeration for $m \leq 5$ and satisfies the construction lower and theorem upper bounds at every archived point. At $m = 2048$, the largest exact family ratio is 5.405. Fixed and shrinking annulus schedules illustrate the regimes of Theorem 3.2. This calculation uses $d = m + 1$ and verifies the constructed class, not trained-model or fixed-width behavior (Appendix I.3).

E2: no amplification in the sampled random families. Let $\hat{c}_s$ be the minimum radius over the evaluated sequences and finite family at stage s . We estimate the dimensionless diagnostic

$$\hat{D}_s = \sqrt{\hat{c}_s^2 + \nu} \frac{\mathfrak{R}_m^{\text{vec}}(\mathcal{N} \circ \mathcal{H}_s^{\text{fin}})}{\mathfrak{R}_m^{\text{vec}}(\mathcal{H}_s^{\text{fin}})}. \quad (23)$$

Thus one is the threshold relative to the sampled-floor Lipschitz scale, not a statement that normalization decreases unscaled vector lengths. The frozen control replaces the denominator by the reference model’s per-example denominator. Figure 1 shows both architecture variation and the baseline stage profile.

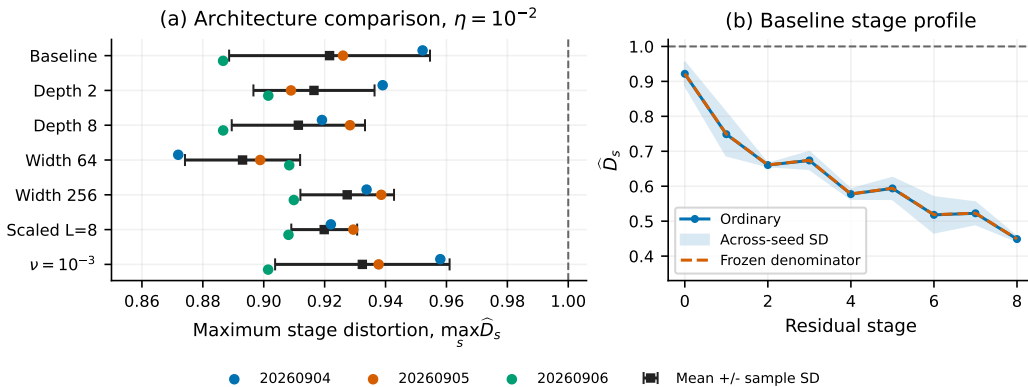


Figure 1: E2 at $m = 512, \eta = 10^{-2}$. Left: each colored point is one seed’s maximum over stages; black bars show mean $\pm$ sample SD across three seeds. Right: baseline stage means and an across-seed SD band; ordinary and frozen curves nearly overlap. These bands are not the pointwise sign-bootstrap intervals.

All 612 estimates lie below one, with range $[0.3422, 0.9594]$, and every pointwise sign-bootstrap upper endpoint is below one; the largest is 0.9602. Each of the 60 condition maxima occurs at stage

zero. The baseline maximum is 0.922 ± 0.033 , while its final-stage diagnostic is 0.449 ± 0.006 . The largest absolute ordinary–frozen point difference over all stage rows is 0.001446. This is a small observed difference; the compact archive does not contain the per-sign maxima needed for a paired-difference interval or an equivalence claim.

The qualitative observation is stable, but architecture rankings are not. For example, width 64 minus baseline changes from -0.0803 to $+0.0217$ between the first and third seeds. Increasing η tenfold slightly decreases the maximum diagnostic in all 21 seed–architecture pairs. These observations do not support a monotone causal relation between radius spread and distortion. The earlier 128-direction baseline also showed no amplification across five scales; its distinct Monte Carlo protocol is reported separately in Appendix I.4.

E3: the uniform recursion is conservative in every seed. No candidate floor in $\{0.25, 0.5, 0.75\}$ times the empirical reference profile passes the sufficient test in any of the 42 distinct seed–architecture–scale settings. E3 uses the complete 512-sequence fit set even in the additional baseline m rows; these are not extra floor tests. Failure does not establish that the actual model loses its floor. The unconditional zero-floor recursion remains available, but its final displacement-to-observation ratio ranges from 1.87×10^{18} to 2.81×10^{80} .

Depth makes the mechanism visible: the mean $\log_{10} \Pi_0$ at $L = 2, 4, 8$ is respectively 20.80 ± 0.11 , 41.43 ± 0.13 , and 82.99 ± 0.02 (Figure 2). Over these tested depths, near-linearity of the logarithm corresponds to exponential growth of this product, not an empirical lower bound on generalization error. Scaling $L = 8$ reduces the log-product by 9.60 decades on average and measured final displacement by 70–82% in matched seeds. Its mean log-slack is still 71.38. Raising ν to 10^{-3} reduces the baseline log-product by 7.98 decades on average, but also fails to certify a positive floor.

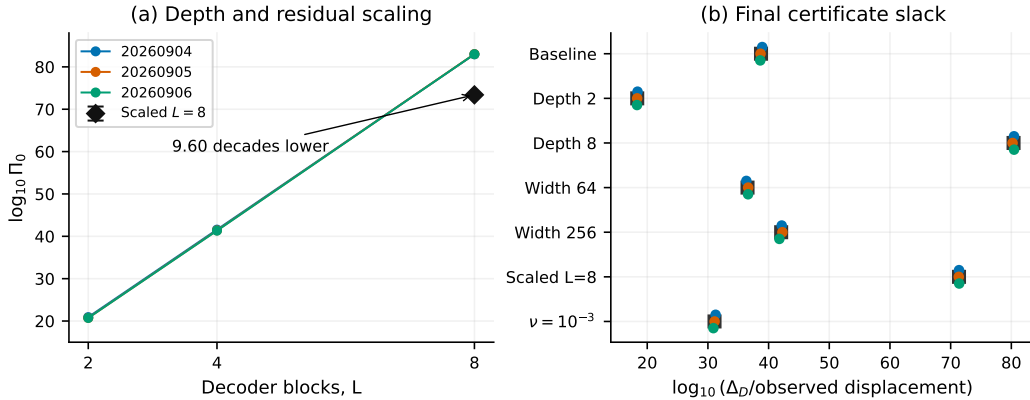


Figure 2: E3 at $m = 512, \eta = 10^{-2}$. Left: $\log_{10} \Pi_0$ versus depth for each seed; curves almost coincide. The scaled depth-8 model lowers the product but remains far above useful values. Right: $\log_{10}$ of final certificate slack, with seed points and mean $\pm$ sample SD. Logs are taken before averaging.

E4: localization improves constants without a useful bound. Let U_N denote the numerically minimized expression inside (21), before its cap at $B/2$. Table 1 compares matched local and global budgets. At $m = 512$, $U_N/(B/2)$ ranges from 232 to 498 across both scales and all seeds. Every one of the 60 condition rows therefore reports the full range $B = 21.94591$ as its final gap bound. The mean baseline capped test-minus-fit gap is 0.00576 ± 0.00484 ; it is descriptive and is not a test of the theorem’s uniform coverage.

Global-to-local K ratios range from 64 to 97 at $\eta = 10^{-2}$ and 629 to 957 at $\eta = 10^{-3}$, yet raw-objective improvement is only 1.17–6.63%. The archived coefficients satisfy a sufficient derivative test ensuring that the raw objective is minimized at $\alpha = B$ (Lemma I.1). The stage-integral term then vanishes, leaving $4B$ and the embedding entropy term, whose A_{loc} already contains Π_0 . The large reduction of K alone consequently does not imply a comparable reduction of this optimized

Configuration	$K_{\text{glob}}/K_{\text{loc}}$	$U_N/(B/2)$	Gain (%)	$10^3 \widehat{\text{gap}}$
Baseline	87.2 ± 1.8	339.6 ± 0.5	2.279 ± 0.007	5.76 ± 4.84
Depth 2	79.3 ± 2.1	243.1 ± 0.7	4.347 ± 0.023	6.79 ± 3.54
Depth 8	90.5 ± 1.5	477.4 ± 0.1	1.165 ± 0.001	7.70 ± 5.05
Width 64	92.8 ± 3.8	235.8 ± 0.5	2.385 ± 0.010	5.99 ± 4.48
Width 256	65.1 ± 1.2	497.1 ± 0.5	2.117 ± 0.005	6.41 ± 5.16
Scaled L=8	88.3 ± 2.8	449.5 ± 0.1	1.313 ± 0.001	6.93 ± 4.46
$\nu = 10^{-3}$	84.6 ± 1.8	301.6 ± 0.7	2.873 ± 0.012	6.63 ± 3.43

Table 1: E4 at $m = 512, \eta = 10^{-2}$; mean $\pm$ sample SD across three seeds. Gain is $100(1 - U_{\text{local}}/U_{\text{global}})$. Every final gap bound equals B . The empirical gap is the capped reference test loss minus fit loss, not a worst-case class gap.

objective. The three-seed baseline sample-size curve lowers $U_N/(B/2)$ from about 946 at $N = 64$ to 340 at $N = 512$ for $\eta = 10^{-2}$, still far above the cap (Appendix I.6).

7 DISCUSSION AND LIMITATIONS

Radius variation identifies a precise obstruction to one vector contraction strategy. The sharp theorem is worst-case over dimensions and evaluation classes. Its learned-embedding realization establishes architectural compatibility, not typical behavior after training. A finite set of perturbed checkpoints can test whether the effect is visible, but cannot certify the supremum over a continuous parameter class.

The empirical findings sharpen these distinctions. Non-amplification under the sampled-floor diagnostic holds throughout the three-seed architecture study, whereas the explicit radius recursion is many orders of magnitude above measured finite-family displacement and all reported loss bounds are vacuous. Architectural changes alter the severity of the latter finding consistently, but the ordering of E2 distortion across architectures can change with the seed. Thus the evidence supports persistence of the qualitative observation, not a universal ranking or a claim that normalization is generally nonexpansive.

The experiment uses a controlled synthetic task, three seeds, and 32 random directions per architecture (128 for the earlier single-reference baseline). The vector process is nonzero even for a singleton class, so small-neighborhood ratios may be dominated by the reference contribution. Neither a finite-family maximum nor a numerical fit-set minimum certifies a continuous neighborhood or population radius floor. Pointwise sign-bootstrap intervals are conditional on the sampled family and do not provide a simultaneous confidence statement.

Adversarial-family search evaluated on fresh signs, matched family-size checks, full-support small-task radius measurements, and document-disjoint text data remain the next tests. Jacobian and trajectory-local diagnostics could identify structure suppressed by the product recursion, but replacing uniform bounds by sampled derivatives would require a new proof. The present results establish a sharp obstruction and a valid, explicitly measured route to a loss bound; they do not establish that the current certificate is tight or that useful transformer generalization guarantees are impossible.

AI USE STATEMENT

Generative AI (OpenAI ChatGPT/Codex) was used to restructure the manuscript, review and formulate mathematical claims, develop proof arguments, check related literature, design and implement the experimental protocols, analyze the archived run outputs, and prepare the $\text{\LaTeX}$ sources. Training and experimental evaluation were executed by the researcher in Google Colab; reported measurements are taken from those archived outputs. Final independent author verification and approval are still pending.

REPRODUCIBILITY STATEMENT

All theoretical statements include their assumptions and proofs in the appendix. The source package includes the official ICLR 2027 style files, bibliography, build instructions, experiment code, input result archives, and a script that audits the archives and regenerates the three-seed figures, tables, and CSV summaries. Appendix I distinguishes completed protocols from outstanding experiments and states the limits of the archived outputs. No additional model training was performed during manuscript integration.

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A PROOFS FOR NORMALIZATION AND SHARPNESS

A.1 PROOF OF THEOREM 3.1

Proof. Let $\tilde{r}_i = \max\{r_i^0, c\}$ and $q_i^0 = (\tilde{r}_i^2 + \nu)^{-1/2}$. Projection onto $[c, \infty)$ cannot increase the distance from an admissible radius. Moreover

$$\left| \frac{d}{dr} (r^2 + \nu)^{-1/2} \right| = \frac{r}{(r^2 + \nu)^{3/2}} \leq b^2 \quad (r \geq c).$$

Consequently $0 \leq q_i^0 \leq b$ and $|q_i(\theta) - q_i^0| \leq \rho b^2$. Decompose $q_i(\theta)h_i(\theta) = q_i^0h_i(\theta) + [q_i(\theta) - q_i^0]h_i(\theta)$. Fixed-coordinate scalar contraction in the dual projection class bounds the first signed-sum complexity by $b\mathfrak{R}_m^{\text{vec}}(\mathcal{H})$. The triangle inequality bounds the second by $M\rho b^2$. The gain contributes at most G . $\square$

For clarity, the fixed-coordinate contraction used here is scalar contraction on the class of projections $\{(\langle u, h_i(\theta) \rangle)_i : \|u\|_2 \leq 1, \theta \in \Theta\}$. Its scalar complexity is exactly (2). Coefficients q_i^0/b_c are fixed and lie in $[0, 1]$, so they cannot increase this complexity. The reference profile may depend on the fixed evaluation inputs; it must be common across hypotheses. This deterministic statement alone supplies no high-probability claim for a data-selected class.

A.2 A MOMENT INEQUALITY

Lemma A.1 (An elementary Hilbert-space lower bound). *For deterministic vectors $v_1, \dots, v_m$ in a Hilbert space,*

$$\frac{1}{\sqrt{3}} \left(\sum_i \|v_i\|_2^2 \right)^{1/2} \leq \mathbb{E}_\varepsilon \left\| \sum_i \varepsilon_i v_i \right\|_2 \leq \left(\sum_i \|v_i\|_2^2 \right)^{1/2}.$$

Proof. Let $X = \left\| \sum_i \varepsilon_i v_i \right\|_2$ and $A = \left(\sum_i \|v_i\|_2^2 \right)^{1/2}$. Then $\mathbb{E}X^2 = A^2$ and $\mathbb{E}X^4 = A^2 + 4 \sum_{i < j} \langle v_i, v_j \rangle^2 \leq 3A^2$. Hölder's inequality gives $\mathbb{E}X^2 \leq (\mathbb{E}X)^{2/3}(\mathbb{E}X^4)^{1/3}$, proving the lower bound. The upper bound is Jensen's inequality. $\square$

A.3 PROOF OF THEOREM 3.2

One valid choice of constants is

$$C_{\nu, \rho_0} = \frac{\sqrt{3}(1 + 2\rho_0)}{\sqrt{1 + \nu}}, \quad c_{\nu, \rho_0} = \frac{\sqrt{1 + \nu}}{(1 + 2\rho_0)[1 + \nu + 4\rho_0(1 + \rho_0)]^{3/2}}. \quad (24)$$

Proof. Fix any class in (5). For a fixed member of the class, $\sum_i \|h_i\|_2^2 \geq md$, and Lemma A.1 yields $\mathfrak{R}_m^{\text{vec}}(\mathcal{H}) \geq \sqrt{d/(3m)}$. The profile inequality at reference radius $1 + \rho$, radius floor one, and Euclidean envelope $\sqrt{d}(1 + 2\rho)$ gives

$$\frac{\sqrt{1 + \nu} \mathfrak{R}_m^{\text{vec}}(\mathcal{N} \circ \mathcal{H})}{\mathfrak{R}_m^{\text{vec}}(\mathcal{H})} \leq 1 + \frac{\sqrt{3}(1 + 2\rho)\rho\sqrt{m}}{\sqrt{1 + \nu}}.$$

This proves the upper bound.

For the binary class (7), orthogonality gives

$$\left\| \sum_i \varepsilon_i h_i(a) \right\|_2^2 = d \left(S^2 + \sum_i a_i^2 \right).$$

The unnormalized supremum is attained by taking every $a_i = t$, which gives (8); Jensen's inequality uses $\mathbb{E}S^2 = m$. For each sign vector choose $a_i = 0$ if $\varepsilon_i = 1$ and $a_i = t$ otherwise. The normalized sum has e_0 projection $\sqrt{d}(q_0 n_+ - q_1 n_-)$, where $q_0 = (1 + \nu)^{-1/2}$ and $q_1 = (1 + t^2 + \nu)^{-1/2}$. Since $\mathbb{E}n_+ = \mathbb{E}n_- = m/2$, its expected projection is $m\sqrt{d}(q_0 - q_1)/2$. The supremum norm is at least this

projection for each sign vector, even when the projection is negative. Hence the normalized-to-raw ratio in (5) is at least

$$\frac{\sqrt{m}}{2\sqrt{1+t^2}} \left(1 - \sqrt{\frac{1+\nu}{1+t^2+\nu}} \right). \quad (25)$$

Take $t^2 = 4\rho(1+\rho)$, so $\sqrt{1+t^2} = 1+2\rho$. Writing $A = 1+\nu$ and $T_0 = 4\rho_0(1+\rho_0)$, integration gives

$$1 - \sqrt{\frac{A}{A+t^2}} = \int_0^{t^2} \frac{\sqrt{A}}{2(A+u)^{3/2}} du \geq \frac{t^2\sqrt{A}}{2(A+T_0)^{3/2}}.$$

Use $t^2 \geq 4\rho$ and $\sqrt{1+t^2} \leq 1+2\rho_0$ in (25) to obtain the stated lower constant. A singleton class with every radius one gives distortion exactly one for all m . This also proves the result when $\rho = 0$. Finally, $\max\{1, cx\} \geq \min\{1, c\}(1+x)/2$ for $x \geq 0$, which gives the claimed order. The uniform boundedness equivalence follows directly from the two inequalities. $\square$

A.4 DECODER REALIZATION AND LAYERNORM

Proposition A.2 (Realization by learned token embeddings). *For $V \geq \max\{m, 2\}$, the class in (7) is exactly the hidden-state evaluation class of a length-one decoder subclass on m distinct input tokens, with embedding RMS row budget $\sqrt{1+t^2}$, gain one, and all residual branches set to zero. Every hidden-state radius on the support of these tokens is at least one.*

Proof. Restrict embedding row i to the values $\sqrt{d}(e_0 + a_i e_i)$, where $a_i \in \{0, t\}$, and fix any unused rows to zero. Each residual layer is the identity when its branch weights are zero, so the hidden states remain these embedding rows at every depth. All upper bounds on branch matrix norms admit the zero matrices. $\square$

The realization allows d and V to grow with m and constrains embedding rows in RMS units. Their Euclidean budget is consequently $\sqrt{d}\sqrt{1+t^2}$. It does not establish the same obstruction for fixed embeddings, a dimension-independent Euclidean row budget, or the capped loss of a complete decoder. In particular, the realization rules out a contraction asserted uniformly for every such decoder subclass. It does not prove failure for the entire class defined only by norm upper bounds: the ratio of two Rademacher complexities is not monotone when the class is enlarged.

Corollary A.3 (Mean-zero LayerNorm realization). *The construction in (7) applies to gain-one LayerNorm with regularizer ν when $e_0, \dots, e_m$ are chosen in the mean-zero subspace of $\mathbb{R}^{m+2}$ and the RMS scaling uses $d = m+2$.*

Proof. The subspace has dimension $m+1$. All hidden vectors have zero coordinate mean, so centering leaves them unchanged. LayerNorm and RMS normalization therefore coincide on every constructed vector. $\square$

A.5 WHY VARIATION ALONE DOES NOT FORCE LARGE DISTORTION

Remark A.4 (Radius variation need not cause large distortion). For this remark only, let $\mathcal{Q} = \{(\|h_i(\theta)\|_0^2 + \nu)^{-1/2}\}_{i=1}^m : \theta \in \Theta\}$. Let $d \geq m$ and $h_i(a) = \sqrt{d}a_i e_i$, with independent $a_i \in [1, 2]$. Then

$$\mathfrak{R}_m^{\text{vec}}(\mathcal{H}) = 2\sqrt{d/m}, \quad \mathfrak{R}_m^{\text{vec}}(\mathcal{N} \circ \mathcal{H}) = \frac{2}{\sqrt{4+\nu}}\sqrt{d/m},$$

whereas $\hat{\mathfrak{R}}_m(\mathcal{Q}) = \frac{1}{2}[(1+\nu)^{-1/2} - (4+\nu)^{-1/2}] > 0$. Indeed the norm is maximized by $a_i = 2$, and the scalar inverse-radius supremum selects its larger value for positive signs and its smaller value for negative signs. Hence a nonvanishing inverse-radius complexity does not itself force a large contraction factor.

B PROOFS FOR THE LOCAL RADIUS CERTIFICATE

B.1 PROOF OF LEMMA 4.1

Proof. The output radius is $\phi(r) = r / \sqrt{\max\{r^2, c^2\} + \nu}$. The function is continuous, vanishes at zero, and has derivative b_c below c and $\nu(r^2 + \nu)^{-3/2} \leq b_c$ above c . Also $\phi(r) \leq b_c r$. If two inputs have radii r, s and angular cosine u , their squared output distance in RMS units is

$$[\phi(r) - \phi(s)]^2 + 2\phi(r)\phi(s)(1 - u) \leq b_c^2[(r - s)^2 + 2rs(1 - u)].$$

This proves the global Lipschitz bound, including zero inputs by continuity. The agreement and output-radius assertions follow directly from the formula. $\square$

The argument proves a two-point Lipschitz estimate on the whole space. A derivative bound restricted to the exterior of a ball would not by itself justify integration along a segment crossing that ball. The extension removes precisely this issue, and $\nu > 0$ handles $c = 0$ without a singular limit.

B.2 PROOF OF THEOREM 4.2

Proof. Fix the proposed floor sequence and replace the normalizer before branch s by $\overline{\mathcal{N}}_{c_{s-1}}$. Denote the resulting streams by $\overline{H}^s(\theta)$. The reference satisfies $a_s \geq c_s$ on $\mathcal{X}$, so induction gives $\overline{H}^s(\theta^0) = H^s(\theta^0)$ there. Write

$$e_s = \sup_{\theta \in \Theta_{\text{loc}}, X \in \mathcal{X}} \left\| \overline{H}^s(X; \theta) - H^s(X; \theta^0) \right\|_{\infty, \circ}.$$

At initialization $e_0 \leq e_E$. Both normalized streams have RMS radius at most one. For fixed θ, X , split the branch difference as

$$\begin{aligned} & F_s(\overline{\mathcal{N}}_{c_{s-1}}(\overline{H}^{s-1}(\theta)); W_s) - F_s(\overline{\mathcal{N}}_{c_{s-1}}(H^{s-1}(\theta^0)); W_s^0) \\ &= F_s(\overline{\mathcal{N}}_{c_{s-1}}(\overline{H}^{s-1}(\theta)); W_s) - F_s(\overline{\mathcal{N}}_{c_{s-1}}(H^{s-1}(\theta^0)); W_s) \\ &\quad + F_s(\overline{\mathcal{N}}_{c_{s-1}}(H^{s-1}(\theta^0)); W_s) - F_s(\overline{\mathcal{N}}_{c_{s-1}}(H^{s-1}(\theta^0)); W_s^0). \end{aligned}$$

The first difference is at most $\Lambda_s \omega_{s-1} e_{s-1}$. For the second, telescope one matrix at a time. At every replacement the linear input has Euclidean row norm at most b_j , so its linear output changes by at most $b_j \tau_j$. The stage certificate gives branch error at most $\ell_j b_j \tau_j$. The ambient caps hold for all the hybrid matrices, even when a coupling among the original parameters has been relaxed. Summing the errors yields q_s . Including the residual connection,

$$e_s \leq (1 + \lambda_s \Lambda_s \omega_{s-1}) e_{s-1} + \lambda_s q_s,$$

whence $e_s \leq \Delta_s$ by induction.

For every position, reverse triangle now gives

$$\left\| \overline{H}_t^s(X; \theta) \right\|_{\circ} \geq \left\| H_t^s(X; \theta^0) \right\|_{\circ} - \Delta_s \geq a_s - \Delta_s \geq c_s.$$

This is established for the extended network before any claim of agreement for the candidate. Agreement with the original candidate then follows by induction, beginning with the shared embedding stream. Thus the stream estimate holds for original networks. A second application of reverse triangle gives the radius-profile estimate.

If every $c_s = 0$, then $\overline{\mathcal{N}}_0 = \mathcal{N}$ everywhere. The error recursion is valid without any feasibility test, and reverse triangle with nonnegativity of the radius gives the stated floor. $\square$

The recursion has the explicit solution

$$\Delta_s = e_E \prod_{u=1}^s L_u + \sum_{v=1}^s \lambda_v q_v \prod_{u=v+1}^s L_u. \quad (26)$$

The empty product equals one. A proposed positive-floor certificate can be checked by finite arithmetic once its ambient caps and reference floors are known. If a proposed certificate fails, it gives no conclusion of that form; the zero-floor calculation remains valid. One may obtain floors from the zero-floor calculation and recompute a candidate certificate, but every improved candidate must satisfy its own displayed feasibility inequalities.

B.3 PROOF OF COROLLARY 4.3

Proof. Every branch vanishes at zero and has Lipschitz constant Λ_s on the unit RMS ball. Its normalized input is in that ball, so $\|H^s\|_{\infty,0} \leq B_0 + \sum_{u=1}^s \lambda_u \Lambda_u = B_s$. Apply Theorem 3.1 with $M = \sqrt{d}B_s$, $\rho = \Delta_s$, $b_c = \omega_s$, and reference profile given by $H^s(\theta^0)$ at the selected evaluations. Any fixed member of the hidden-state class has all radii at least c_s , so Lemma A.1 gives $\mathfrak{R}_m^{\text{vec}}(\mathcal{H}_s) \geq \sqrt{dc_s}/\sqrt{3m}$. Dividing the additive term by $\omega_s \mathfrak{R}_m^{\text{vec}}(\mathcal{H}_s)$ proves (17). If the specified perturbation budgets scale by η , linearity of (14) gives $\Delta_s = \eta D_s$, where $D_0 = e$ and $D_s = L_s D_{s-1} + \lambda_s \sum_{j:s(j)=s} \ell_j b_j t_j$. The claimed sufficient scale follows for fixed certified floors and caps. $\square$

B.4 A NONEMPTY NEIGHBORHOOD WITH A POPULATION CERTIFICATE

The certificate is not limited to an assumed trained-model floor. Fix any embedding table E^0 with every row of RMS radius $a > 0$, set all reference branch matrices to zero, and take any reference readout within its cap. Every reference stream has radius a on every token sequence, so $a_s = a$ is known without a distributional assumption. Take $B_0 \geq a$ and choose the candidate floors $c_s = a/2$. Put $b = ((a/2)^2 + \nu)^{-1/2}$ and $Q = \sum_s \lambda_s \Lambda_s$. Equation (26) and $1 + x \leq e^x$ imply

$$\Delta_s \leq e^{bQ} \left(e_E + \sum_{v=1}^D \lambda_v q_v \right) \quad (0 \leq s \leq D).$$

Consequently the explicit sufficient condition

$$e_E + \sum_{v=1}^D \lambda_v \sum_{j:s(j)=v} \ell_j b_j \tau_j \leq \frac{a}{2} e^{-bQ} \quad (27)$$

certifies all floors $a/2$. With positive ambient budgets this permits nonzero branch perturbations, and the reference itself is always included. This example proves that the positive-floor local theorem has concrete nonempty instances. It does not establish a useful certificate size for trained language models; the right side can be small at large depth.

B.5 FROM RAW WEIGHTS AND GAINS TO EFFECTIVE BUDGETS

Suppose an effective first matrix is $A = \text{diag}(\gamma)W$, with reference $A^0 = \text{diag}(\gamma^0)W^0$. If $\|\gamma\|_\infty \leq G$ and $\|\gamma - \gamma^0\|_\infty \leq e_\gamma$, then

$$\begin{aligned} \|A - A^0\|_{\text{op}} &\leq G\|W - W^0\|_{\text{op}} + e_\gamma\|W^0\|_{\text{op}}, \\ \|A - A^0\|_{2,1} &\leq G\|W - W^0\|_{2,1} + e_\gamma\|W^0\|_{2,1}. \end{aligned} \quad (28)$$

Indeed, $A - A^0 = \text{diag}(\gamma)(W - W^0) + \text{diag}(\gamma - \gamma^0)W^0$; apply the triangle inequality and the operator norm of a diagonal matrix, column by column for the mixed norm. Thus raw parameter neighborhoods imply explicit effective budgets in (11). Output matrices do not absorb a second gain.

For a tied readout $C = \text{diag}(\gamma_f)E^\top$ with a correspondingly tied reference, bounds $\|\gamma_f\|_\infty \leq G_f$, $\|\gamma_f - \gamma_f^0\|_\infty \leq e_f$, and $\max_v \|E_v^0\|_0 \leq B_0^0$ imply

$$\max_v \|C_v - C_v^0\|_2 \leq G_f \sqrt{d} e_E + e_f \sqrt{d} B_0^0. \quad (29)$$

Use this as e_C and the total cap $r_{\text{out}} = G_f \sqrt{d} B_0$. The independent readout cover contains the tied class and is an upper-bound relaxation; it is not a claim of a sharp capacity cost for tying.

C THE DECODER CLASS AND ITS CONSTRAINTS

This section defines the ambient global class. Local classes intersect its individual matrix and embedding constraints with the reference-centered budgets in (11); the translated local sets need not contain zero.

C.1 RESIDUAL STREAM, GAINS, AND THE OUTPUT LOSS

The vocabulary has size $V \geq 2$. Inputs $X = (X_1, \dots, X_T)$ are token sequences, and targets $Y = (Y_1, \dots, Y_T)$ lie in the same vocabulary. An embedding matrix $E \in \mathbb{R}^{V \times d}$ initializes $H_t^0 = E_{X_t, \cdot}$, with

$$\max_{v \leq V} \|E_{v, \cdot}\|_o \leq B_0.$$

There are $D = 2L$ residual sublayers, alternating attention and gated feedforward branches:

$$H^s = H^{s-1} + \lambda_s F_s(\mathcal{N}(H^{s-1})), \quad s = 1, \dots, D. \quad (30)$$

The coefficients $\lambda_s \geq 0$ are fixed, with $\lambda_s = 1$ giving the usual unscaled residual update. Maps act rowwise except attention. There are no additive biases, dropout variables, or learned position maps in the class analyzed here.

Learned normalization gains are included by an exact reparameterization. For an attention gain γ_s replace the first matrices by

$$A_Q = \text{diag}(\gamma_s)W_Q, \quad A_K = \text{diag}(\gamma_s)W_K, \quad A_V = \text{diag}(\gamma_s)W_V.$$

For a gated branch use $A_g = \text{diag}(\gamma_s)W_g$ and $A_u = \text{diag}(\gamma_s)W_u$. The output matrices W_O, W_d are unchanged. A raw bound $\|W\|_{2,1} \leq R$, $\|W\|_{\text{op}} \leq S$ and $\|\gamma_s\|_\infty \leq G_s$ implies effective bounds $G_s R, G_s S$ on these first matrices. The gain is absorbed once.

The final logits are

$$z_t = \mathcal{N}(H_t^D)C, \quad \max_{v \leq V} \|C_{\cdot v}\|_2 \leq r_{\text{out}}. \quad (31)$$

For a tied readout, $C = \text{diag}(\gamma_f)E^\top$; thus $r_{\text{out}} = G_f \sqrt{d}B_0$ is valid when $\|\gamma_f\|_\infty \leq G_f$. We cover the larger class in which effective matrices and C vary independently within their constraints. This relaxation contains the original gained and tied model, so upper bounds apply to it. It does not preserve lower bounds for independently varying readouts.

Fix a cap parameter $c_{\log} > 0$ and apply $\psi(x) = c_{\log} \tanh(x/c_{\log})$ to every logit. The sequence loss is

$$g_\theta(X, Y) = \frac{1}{T} \sum_{t=1}^T \left[\log \sum_{v=1}^V e^{\psi(z_{t,v})} - \psi(z_{t,Y_t}) \right]. \quad (32)$$

The cap is fixed and 1-Lipschitz. Cross-entropy is 2-Lipschitz in the maximum logit norm, since its gradient has ℓ_1 norm at most two. In particular

$$0 \leq g_\theta \leq B, \quad B = \log(1 + (V-1)e^{2c_{\log}}). \quad (33)$$

No independence between the T summands is assumed.

Assumption C.1 (Fixed constraints and population floors). All effective matrix norm bounds, B_0 , r_{out} , and λ_s are deterministic and fixed before the training sample is observed. Each matrix belongs to a nonempty bounded constraint set containing zero, with a specified mixed-norm budget R_j and the spectral or column bounds stated below. For a deterministic profile $c_0, \dots, c_D \geq 0$, the original model class is restricted to parameters satisfying

$$\|H_t^s(X; \theta)\|_o \geq c_s \quad \text{for all } s, t, \theta \text{ and every } X \text{ in the population input support.} \quad (34)$$

The choice $c_s = 0$ is always valid and covers the complete bounded parameter class. Positive floors give a potentially smaller class and stronger constants. They are additional assumptions, not consequences of normalization or empirical minimum radii.

For the proof, replace the normalizer before branch s by $\overline{\mathcal{N}}_{c_{s-1}}$ and the final normalizer by $\overline{\mathcal{N}}_{c_D}$. Remove the hidden-state floor restrictions from this *extended* parameter class. Induction on s shows that every original model satisfying (34) agrees with its extended counterpart on the population support. Every extended normalized stream has RMS radius at most one.

C.2 GROUPED-QUERY ATTENTION

We use the grouped-query parameterization of Ainslie et al. (2023) and permit fixed orthogonal rotary maps as in Su et al. (2021).

There are H query heads, J key/value groups, $H = gJ$, and $d = Hd_h$. The group of head h is $\mu(h)$, with exactly g heads in each group. Write

$$A_Q \in \mathbb{R}^{d \times d}, \quad A_K, A_V \in \mathbb{R}^{d \times Jd_h}, \quad W_O \in \mathbb{R}^{d \times d}.$$

Subscripts h or μ indicate column blocks of width d_h . At a normalized input Z , head h has queries $ZA_{Q,h}$, keys $ZA_{K,\mu(h)}$, and values $ZA_{V,\mu(h)}$. Fixed orthogonal rotary maps may be applied to queries and keys at each position.

For each query position i , a deterministic nonempty set $\mathcal{J}_i \subseteq \{1, \dots, T\}$ specifies visible keys. The logits on that set are the rotary query–key inner products divided by $\sqrt{d_h}$, optionally followed by a fixed scalar 1-Lipschitz score cap. Rowwise softmax gives probabilities $P_h(i, j)$. With $U_h = P_h ZA_{V,\mu(h)}$, the branch is

$$F_{\text{att}}(Z) = [U_1 | \dots | U_H] W_O.$$

The causal mask is a special case. The normalizer in each attention row is taken only over its visible keys.

Fix class-wide bounds

$$\max_h \|A_{Q,h}\|_{\text{op}} \leq q, \quad \max_{\mu} \|A_{K,\mu}\|_{\text{op}} \leq k, \quad \|A_{V,\mu}\|_{\text{op}} \leq v_{\mu}, \quad \|W_O\|_{\text{op}} \leq S_O.$$

Define

$$\beta_Q = \sqrt{H}q, \quad \beta_K = \sqrt{H}k, \quad \beta_V = \sqrt{H} \max_{\mu} v_{\mu}, \quad \Sigma_V = \left(g \sum_{\mu=1}^J v_{\mu}^2 \right)^{1/2}. \quad (35)$$

For $\|Z\|_{\infty,0} \leq 1$, a query or key head has Euclidean row radius at most $\sqrt{d_h}\beta_Q$ or $\sqrt{d_h}\beta_K$; a value head has radius at most $\sqrt{d_h}\beta_V$. All quantities can depend on the sublayer index.

C.3 GATED FEEDFORWARD BRANCHES

For width w the branch is

$$F_{\text{ff}}(Z) = [\text{swish}(ZA_g) \odot (ZA_u)]W_d, \quad \text{swish}(x) = \frac{x}{1 + e^{-x}},$$

with $A_g, A_u \in \mathbb{R}^{d \times w}$ and $W_d \in \mathbb{R}^{w \times d}$. Fix spectral caps S_g, S_u, S_d and column caps $C_g \geq \max_j \|(A_g)_{\cdot j}\|_2$, $C_u \geq \max_j \|(A_u)_{\cdot j}\|_2$. One may always take $C_g = S_g$, $C_u = S_u$. Put

$$\beta_a = \sqrt{d}C_g, \quad \beta_b = \sqrt{d}C_u, \quad \kappa_{\text{sw}} = 1 + e^{-1}.$$

This explicit choice bounds $\sup_x |\text{swish}'(x)|$: with $\sigma(x) = (1 + e^{-x})^{-1}$, $\text{swish}'(x) = \sigma(x) + x\sigma(x)(1 - \sigma(x))$ and $|x\sigma(x)(1 - \sigma(x))| \leq |x|e^{-|x|} \leq e^{-1}$. We do not need the optimal scalar Lipschitz constant.

D BRANCH AND PARAMETER PERTURBATION CERTIFICATES

Two different constants are needed. A branch Lipschitz constant propagates an error in its input stream. A stage sensitivity propagates an approximation error in the output of one linear map, with that map’s input held fixed. The latter is measured in maximum Euclidean row norm, whereas branch outputs are measured in maximum RMS row norm.

Lemma D.1 (Softmax on a fixed visible set). *For any two finite score vectors on the same nonempty visible set,*

$$\|\text{softmax}(a) - \text{softmax}(b)\|_1 \leq \|a - b\|_{\infty}.$$

Proof. At a probability vector p , the derivative applied to u is $p \odot (u - \langle p, u \rangle \mathbf{1})$. Its ℓ_1 norm is the mean absolute deviation of a random variable taking values u_j with probabilities p_j . If $\|u\|_\infty \leq 1$, this is at most its standard deviation, which is at most one. Integrate the derivative along the segment between the score vectors. $\square$

Proposition D.2 (Branch Lipschitz constants). *On $\{Z : \|Z\|_{\infty, \circ} \leq 1\}$ the following are valid Lipschitz certificates in maximum RMS row norm:*

$$\Lambda_{\text{att}} = S_O \left[\Sigma_V + 2\sqrt{d_h} \beta_Q \beta_K \beta_V \right], \quad (36)$$

$$\Lambda_{\text{ff}} = S_d [\kappa_{\text{sw}} \beta_b S_g + \beta_a S_u]. \quad (37)$$

Both branches vanish at zero, so their output RMS radius on this domain is at most the corresponding Λ .

Proof. Let $\delta = \|Z - Z'\|_{\infty, \circ}$. For an attention head, query and key perturbations have Euclidean row norm at most $\sqrt{d_h} \beta_Q \delta$ and $\sqrt{d_h} \beta_K \delta$. The score perturbation is therefore at most $2\sqrt{d_h} \beta_Q \beta_K \delta$ in maximum norm. Rotary maps preserve these norms and a 1-Lipschitz score cap cannot increase the difference. By Lemma D.1, changing the probabilities changes a head output by at most $2d_h \beta_Q \beta_K \beta_V \delta$ in Euclidean norm. Concatenating H heads and dividing by $\sqrt{d}$ gives $2\sqrt{d_h} \beta_Q \beta_K \beta_V \delta$.

When the probabilities are held fixed, the value perturbation of a head in group μ is at most $\sqrt{d} v_\mu \delta$. The concatenated RMS perturbation is at most $(g \sum_\mu v_\mu^2)^{1/2} \delta = \Sigma_V \delta$. Multiplication by W_O completes (36).

For a gated branch write $a = ZA_g$, $b = ZA_u$, and similarly a', b' . At each row,

$$\|\text{swish}(a) \odot b - \text{swish}(a') \odot b'\|_2 \leq \kappa_{\text{sw}} \beta_b \|a - a'\|_2 + \beta_a \|b - b'\|_2.$$

Here $|\text{swish}(x)| \leq |x|$, and the activation bounds hold for both inputs. Apply the spectral caps and divide by $\sqrt{d}$. Zero input gives zero values and zero gate outputs. $\square$

Stage matrix	Input radius b_j	Sensitivity ℓ_j
A_Q	$\sqrt{d}$	$S_O \beta_K \beta_V / \sqrt{H}$
A_K	$\sqrt{d}$	$S_O \beta_Q \beta_V$
A_V	$\sqrt{d}$	$S_O / \sqrt{d_h}$
W_O	$\sqrt{d} \Sigma_V$	$1/\sqrt{d}$
A_g	$\sqrt{d}$	$S_d \kappa_{\text{sw}} \beta_b / \sqrt{d}$
A_u	$\sqrt{d}$	$S_d \beta_a / \sqrt{d}$
W_d	$\sqrt{d} \beta_a S_u$	$1/\sqrt{d}$

Table 2: Stage certificates before multiplication by residual coefficient λ_s . All symbols are class-wide bounds for the containing sublayer, defined in the decoder specification.

Proposition D.3 (Linear stage certificates). *Consider replacing one matrix by another matrix in the same constraint set, with the input to that linear map fixed. Suppose its linear outputs differ by at most η_j in maximum Euclidean row norm. Then the branch output differs by at most $\ell_j \eta_j$ in maximum RMS row norm, with b_j and ℓ_j in Table 2. The number b_j bounds the Euclidean row radius of that stage’s fixed input, for every admissible choice of the other matrices.*

Proof. For a query replacement, let the per-head query errors at a fixed query position be η_h . Since all query heads share this position, $\sum_h \eta_h^2 \leq \eta_Q^2$. The score error in head h is at most $\beta_K \eta_h$, so its value-weighting error is at most $\sqrt{d_h} \beta_V \beta_K \eta_h$. Concatenation, RMS conversion, and W_O give $S_O \beta_K \beta_V \eta_Q / \sqrt{H}$.

For a key replacement, each head score error is at most $\beta_Q \eta_K$. A head’s probability-weighted value error is at most $\sqrt{d_h} \beta_V \beta_Q \eta_K$. Concatenation gives $S_O \beta_Q \beta_V \eta_K$ in RMS norm. In contrast to the

query case, different heads can attain their maximum key error at different positions. No interchange of a maximum over positions and a sum over groups is used.

For a value replacement, every head’s weighted value error has Euclidean norm at most η_V . The concatenated RMS error is $\sqrt{H/d}\eta_V = \eta_V/\sqrt{d_h}$ before W_O . For the output projection itself the RMS conversion contributes $1/\sqrt{d}$. Its input obeys $\| [U_1] \cdots [U_H]_i \|_2 \leq \sqrt{d}\Sigma_V$.

The two gate sensitivities follow from the product estimate in the proof of Proposition D.2. The input of W_d has norm at most $\|\text{swish}(a)\|_\infty \|b\|_2 \leq \beta_a \sqrt{d}S_u$. Again, its output error is converted from Euclidean to RMS units by $1/\sqrt{d}$. $\square$

The sensitivities hold for hybrid choices of matrices: one matrix may already be a covering center while another is an original parameter. This is why the centers in the next section are selected inside the matrix constraint sets.

E PROPER LINEAR COVERS AND PROPAGATION COEFFICIENTS

E.1 A EUCLIDEAN-INPUT LINEAR COVERING LEMMA

We include a direct proof so that no change of input norm is hidden in an imported entropy estimate. Its greedy approximation step is closely related to the online-learning approach to linear covers [Zhang \(2002\)](#); mixed-norm covers also appear in [Edelman et al. \(2022\)](#).

Lemma E.1 (Proper scalar linear cover). *Fix $x_1, \dots, x_n \in \mathbb{R}^p$ with $\|x_i\|_2 \leq b$. For any nonempty set $\mathcal{W} \subseteq \{w : \|w\|_2 \leq R\}$, there is a proper η -cover in the metric $\max_i |\langle x_i, w - w' \rangle|$ satisfying, for every $\eta > 0$,*

$$\log |\mathcal{N}| \leq 8 \left(\frac{Rb}{\eta} \right)^2 \log(2n + 1). \quad (38)$$

Proof. The case $Rb = 0$ is trivial, and an arbitrary admissible center works when $\eta \geq 2Rb$. Otherwise set $\delta = \eta/2$. For a target w in the ball, start at $v_0 = 0$. Whenever some i satisfies $|\langle x_i, w - v_t \rangle| > \delta$, update

$$v_{t+1} = v_t + \frac{\delta}{b^2} \text{sign}(\langle x_i, w - v_t \rangle) x_i.$$

The squared distance to w decreases by more than δ^2/b^2 :

$$\|w - v_{t+1}\|_2^2 = \|w - v_t\|_2^2 - \frac{2\delta}{b^2} |\langle x_i, w - v_t \rangle| + \frac{\delta^2}{b^4} \|x_i\|_2^2 < \|w - v_t\|_2^2 - \frac{\delta^2}{b^2}.$$

The procedure stops after at most $M = \lfloor R^2 b^2 / \delta^2 \rfloor$ updates. Its final vector approximates every evaluation within δ . All possible final vectors are encoded by at most M choices of an index and a sign. Their number is at most $\sum_{j=0}^M (2n)^j \leq (2n + 1)^M$.

These centers need not belong to $\mathcal{W}$. For every center whose δ -ball meets $\mathcal{W}$, choose one representative from that intersection. The resulting proper cover has radius $2\delta = \eta$ and no more centers. The displayed bound follows, with the stated constant allowing harmless endpoint rounding. $\square$

Lemma E.2 (Proper column mixed-norm cover). *Fix inputs as in Lemma E.1 and a nonempty constraint set $\mathcal{M} \subseteq \{W \in \mathbb{R}^{p \times k} : \|W\|_{2,1} \leq R\}$. For every $\eta > 0$ there is a proper cover, in maximum Euclidean output-row norm, with*

$$\log |\mathcal{N}| \leq 256 \left(\frac{Rb}{\eta} \right)^2 [\log(2k) + \log(2n + 1)]. \quad (39)$$

Additional spectral, column, or head-block constraints in $\mathcal{M}$ are preserved by its centers.

Proof. As before, $Rb = 0$ or $\eta \geq 2Rb$ is immediate. Write $q = Rb/\eta > 1/2$. The vector of column norms of a matrix in the mixed-norm ball lies in $\Delta_R = \{t \in \mathbb{R}_+^k : \sum_j t_j \leq R\}$. For any $t \in \Delta_R$, take a random vector equal to Re_j with probability t_j/R and to zero otherwise. An

average of $M = \lceil 16q^2 \rceil$ independent such vectors has mean t and expected squared distance at most R^2/M . Thus the set of all these averages is an $\eta/(4b)$ -net of Δ_R in Euclidean norm. Its logarithmic cardinality is at most $M \log(k+1) \leq 20q^2 \log(k+1)$, and all its members v still satisfy $\sum_j v_j \leq R$.

Given a matrix with column norms t , replace each column's radius by the selected v_j while keeping its direction. If a column is zero, choose any fixed unit direction. The resulting change in any output row is at most $b \|t - v\|_2 \leq \eta/4$. For this fixed v , cover each column ball of radius v_j by Lemma E.1 at accuracy

$$\eta_j = \frac{\eta}{4} \sqrt{\frac{v_j}{R}} \quad (v_j > 0).$$

The zero columns need no cover. The squared output errors sum to at most $\sum_j \eta_j^2 \leq \eta^2/16$, and the logarithmic cardinalities sum to at most $128q^2 \log(2n+1)$. All column centers have norm at most v_j , so the resulting matrix lies in the mixed-norm ball.

We have constructed an $\eta/2$ -cover of the full mixed-norm ball. For each center whose $\eta/2$ -ball meets $\mathcal{M}$, choose a member of that intersection. This doubles the covering radius and preserves every constraint in $\mathcal{M}$. The two entropy costs above are bounded by (39). $\square$

The factor $\log(2n+1)$ is retained. In particular, this proof does not claim sequence-length independence under a Euclidean input bound. Conversely, it needs no extra logarithm of η^{-1} . All constants used below are deliberately unoptimized.

E.2 RESIDUAL PROPAGATION AND OPTIMIZED STAGE ALLOCATION

For sublayer s , let Λ_s be its certificate in Proposition D.2. Define

$$\omega_s = (c_s^2 + \nu)^{-1/2}, \quad L_s = 1 + \lambda_s \omega_{s-1} \Lambda_s, \quad \Pi_s = \prod_{u=s+1}^D L_u, \quad 0 \leq s \leq D. \quad (40)$$

The empty product is one, so $\Pi_D = 1$ and $\Pi_0 = \prod_{u=1}^D L_u$. Here ω_s is a normalization Lipschitz constant, while b_j in Table 2 is a linear-stage input radius.

Index all linear stages of all branches by j , and let $s(j)$ denote the containing sublayer. Put

$$\kappa_j = \lambda_{s(j)} \ell_j R_j b_j, \quad K_B = \left[\sum_j (\Pi_{s(j)} \kappa_j)^{2/3} \right]^{3/2}. \quad (41)$$

Zero terms can be omitted. Let $k_* \geq 2$ be at least the output width of every linear stage, and define

$$\mathcal{L}_{n,k_*} = \log(2k_*) + \log(2n+1).$$

For the complete loss analysis we also require $k_* \geq V$. Let V_{eff} be the number of distinct input tokens in the sample and $P = \sqrt{V_{\text{eff}} d}$.

F ENTROPY INTEGRATION AND THE GLOBAL SPECIALIZATION

F.1 TRUNCATED ENTROPY INTEGRATION

Lemma F.1 (Truncated entropy bound). *For a scalar evaluation class $\mathcal{F} \subseteq [0, B]^N$ with external maximum-norm entropy $H(\epsilon) = \log \mathcal{N}_\infty(\mathcal{F}, \epsilon)$, and any $0 < \alpha \leq B$,*

$$\widehat{\mathfrak{R}}_N(\mathcal{F}) \leq 4\alpha + \frac{12}{\sqrt{N}} \int_\alpha^B \sqrt{H(\epsilon)} d\epsilon. \quad (42)$$

Also $\widehat{\mathfrak{R}}_N(\mathcal{F}) \leq B/2$.

Proof. Clip covering centers to $[0, B]^N$, which cannot increase their errors. Set $\epsilon_j = B2^{-j}$ and use zero as the single level-zero center. Choose maximum-norm covers at the subsequent scales. For

each evaluation vector choose its closest center at each level. Successive centers differ by at most $3\epsilon_j$ in maximum norm, and the number of possible increments has log-cardinality at most $2H(\epsilon_j)$. The exponential-moment bound for a finite set of Rademacher sums gives a contribution at most $6\epsilon_j\sqrt{H(\epsilon_j)/N}$ at level j .

If $\alpha < B/4$, stop at the largest J with $\epsilon_J \geq 2\alpha$. The remaining error is below 4α . Since $\epsilon_j\sqrt{H(\epsilon_j)} \leq 2 \int_{\epsilon_{j+1}}^{\epsilon_j} \sqrt{H(u)} du$, summing the levels proves (42). For $\alpha \geq B/4$, its 4α term alone suffices. Finally, for every sign vector, $\sup_f \sum_i \epsilon_i f_i \leq B|\{i : \epsilon_i = 1\}|$; take expectations and divide by N . $\square$

F.2 THE GLOBAL NORM-CONSTRAINED CLASS

For the class in Assumption C.1, define

$$\Pi_{\text{out}} = 2r_{\text{out}}\sqrt{d}\omega_D, \quad A = 4B_0\Pi_{\text{out}}\Pi_0, \quad (43)$$

$$\kappa_R = 2\sqrt{V}r_{\text{out}}\sqrt{d}, \quad K = \left[\sum_j (\Pi_{\text{out}}\Pi_{s(j)}\kappa_j)^{2/3} + \kappa_R^{2/3} \right]^{3/2}. \quad (44)$$

These coefficients specialize (19) to neighborhoods centered at zero with absolute, rather than local, budgets.

Corollary F.2 (Global norm-class generalization). *For the population-floor class in Assumption C.1, the entropy bound (20), the complexity bound (21), and the generalization bound (22) hold with $A_{\text{loc}} = A$ and $K_{\text{loc}} = K$. If $r_{\text{out}} = 0$, set $A = K = 0$. Fixed embeddings have no embedding entropy term.*

Proof. Set all reference matrices to zero and choose $e_E = B_0$, $R_j^{\text{loc}} = R_j$, and $e_C = r_{\text{out}}$. Make the spectral perturbation constraints redundant with the ambient caps. This is exactly the global parameter class, restricted to any prescribed population floors. Theorem 5.1 allows imposed floors as well as certified ones, so it applies without a positive reference floor. Its proof uses the extended covering class. Substitution gives (43)–(44); the zero cases follow directly. $\square$

For $c_s = 0$ at every stage this covers the complete bounded parameter class. Positive population floors describe a smaller class. They are not consequences of the zero reference. The corollary is a specialization of the same proof, not an additional contraction argument.

F.3 PREDETERMINED CLASS SELECTION

Corollary F.3 (Predetermined countable stratification). *Let $\{\Theta_a : a \in \mathcal{A}\}$ be a countable collection of classes specified before observing the sample, with weights $\pi_a > 0$ and $\sum_a \pi_a \leq 1$. Apply Theorem 5.1 to each class with confidence parameter $\delta\pi_a$. With probability at least $1 - \delta$, all the resulting bounds hold simultaneously.*

Proof. The probability that at least one class bound fails is at most $\sum_a \delta\pi_a \leq \delta$. $\square$

This permits selection among predetermined norm budgets or floor profiles. If each of $D + 1$ floor coordinates is chosen from an M -point grid, a uniform prior costs $(D + 1) \log M$ in the confidence logarithm. Stratifying a floor alone does not account for arbitrary selection of a reference radius profile or an unrestricted data-dependent neighborhood. Membership in a positive-floor class still concerns the population support.

G COVERING LOCAL CLASSES AND PROOF OF THEOREM 5.1

G.1 TRANSLATION PRESERVES THE PROPER LINEAR-COVER BOUNDS

Fix a reference matrix W^0 and a nonempty matrix set $\mathcal{M}$ with $\|W - W^0\|_{2,1} \leq R^{\text{loc}}$. For fixed design rows x_i ,

$$x_i W = x_i W^0 + x_i(W - W^0), \quad x_i(W - W') = x_i\{(W - W^0) - (W' - W^0)\}.$$

Apply Lemma E.2 to $\mathcal{M} - W^0 \subseteq \{U : \|U\|_{2,1} \leq R^{\text{loc}}\}$. Translating its centers back by W^0 gives a proper cover of $\mathcal{M}$ with the same errors and cardinality. Every selected center satisfies the original ambient caps and the local constraints. The set need not contain zero, and no factor of $\|W^0\|_{2,1}$ enters its entropy. The scalar cover in Lemma E.1 has the identical translation property for a readout column. Reference matrices still affect the ambient caps and hence propagation; this is why the theorem does not remove the reference’s size from every coefficient.

For one observed embedding row, its admissible set is contained in a ball of RMS radius e_E centered at E_v^0 . A maximal η_0 -separated subset of this set has at most $(1 + 2e_E/\eta_0)^d$ members, by packing disjoint Euclidean balls of radius $\sqrt{d}\eta_0/2$ inside the translated ball of radius $\sqrt{d}(e_E + \eta_0/2)$. Maximality makes it a proper η_0 -cover. This also preserves a simultaneous absolute row cap. Cover each observed row and fill unobserved rows with their reference rows. The embedding entropy is at most $P^2 \log(1 + 2e_E/\eta_0)$, with $P^2 = V_{\text{eff}}d$. If extra couplings are present, use the row-product relaxation containing the original embedding class.

G.2 THE CONDITIONAL COVER AND ERROR ALLOCATION

Replace the original normalizer by $\bar{\mathcal{N}}_{c_{s-1}}$ before sublayer s and by $\bar{\mathcal{N}}_{c_D}$ at the readout. Keep the local matrix constraints and discard any stream-floor restrictions on the covering class. Every original network satisfying the population floors agrees with its extended counterpart on the support. This agreement follows either from Theorem 4.2 or directly by induction under the imposed floor restrictions. It is not necessary for a selected covering center’s original, unextended network to satisfy those restrictions.

At a node of the covering tree, the previous approximate stream is fixed. Its normalized rows have Euclidean norm at most $\sqrt{d}$. Cover the query, key, and value matrices, or the two first gate matrices, on this fixed design using the translated proper cover. Conditional on these choices, the design for the output matrix W_O or W_d is also fixed and has its certified input radius b_j . Cover the output matrix on that design. The total number of continuations from a node is bounded uniformly in that node. Hence the log-cardinality of the tree is bounded by the sum of the stage log-cardinalities.

To compare original and covered evaluations of one branch, first change the input under the original branch weights, then replace its first matrices, and finally replace its output matrix at the already covered design. The stage certificates hold for all intermediate matrix choices. If E_s is maximum RMS stream error, then

$$E_s \leq L_s E_{s-1} + \lambda_s \sum_{j:s(j)=s} \ell_j \eta_j, \quad E_D \leq \Pi_0 \eta_0 + \sum_j \Pi_{s(j)} \lambda_{s(j)} \ell_j \eta_j.$$

The readout cap and cross-entropy Lipschitz constant propagate the body error by Π_{out} . At the fixed normalized covered final stream, translate and cover the V readout columns at logit accuracy η_C . Their total log-cardinality is at most

$$8V(e_C \sqrt{d}/\eta_C)^2 \log(2NT + 1),$$

and their added loss error is at most $2\eta_C$. Thus total loss error is at most

$$\Pi_{\text{out}} \Pi_0 \eta_0 + \sum_j a_j \eta_j + a_C \eta_C, \quad a_j = \Pi_{\text{out}} \Pi_{s(j)} \lambda_{s(j)} \ell_j, \quad a_C = 2. \quad (45)$$

For active body stages put $t_j = b_j R_j^{\text{loc}}$, and put $t_C = \sqrt{V} e_C \sqrt{d}$ for the readout. Write $w_j = a_j t_j$ and $w_C = a_C t_C$, as in (19). If $S = \sum_j w_j^{2/3} + w_C^{2/3} > 0$, choose

$$\eta_0 = \frac{\epsilon}{2\Pi_{\text{out}}\Pi_0}, \quad \eta_j = \frac{\epsilon w_j^{2/3}}{2S a_j} \quad \text{for every active stage, including the readout.}$$

The stage errors sum to $\epsilon/2$, and

$$\sum_{j \text{ including } C} (t_j/\eta_j)^2 = \frac{4S^3}{\epsilon^2} = \frac{4K_{\text{loc}}^2}{\epsilon^2}.$$

Use the uniform stage entropy constant 256; it also bounds the scalar readout constant. Substitution into the embedding entropy and the stage sum proves (20). Zero-cost stages need no approximation budget. When $S = 0$, only embeddings can contribute; the same bound remains valid. When $r_{\text{out}} = 0$, every readout is zero and the loss is $\log V$, so one center suffices. These cases avoid division by zero in the allocation. All steps apply at every $\epsilon > 0$.

G.3 STATISTICAL ARGUMENT AND REFERENCE SELECTION

Apply Lemma F.1 to the local scalar loss class. Taking square roots of (20), using $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$, bounding the first integrand at α , and integrating $1/\epsilon$ in the second gives

$$\hat{\mathfrak{R}}_N(\mathcal{G}_{\text{loc}}) \leq 4\alpha + \frac{12}{\sqrt{N}} \left[BP\sqrt{\log(1 + A_{\text{loc}}/\alpha)} + 32K_{\text{loc}}\sqrt{\mathcal{L}}\log(B/\alpha) \right].$$

The scalar range bound gives $\hat{\mathfrak{R}}_N \leq B/2$. Infimizing over α proves the bound by Φ_N . For a fixed function class, let $Z = \sup_g (\mathbb{E}g - N^{-1} \sum_i g(X_i, Y_i))$. Symmetrization gives $\mathbb{E}Z \leq 2\mathbb{E}\hat{\mathfrak{R}}_N$. Replacing one entire sequence changes each of Z and $\hat{\mathfrak{R}}_N$ by at most B/N . Applying bounded differences to both quantities with failure probability $\delta/2$ gives

$$Z \leq 2\hat{\mathfrak{R}}_N + 3B\sqrt{\log(2/\delta)/(2N)} \leq 2\Phi_N + 3B\sqrt{\log(2/\delta)/(2N)}.$$

Finally $Z \leq B$ deterministically. This proves (22). The empirical V_{eff} is valid because its entropy bound holds deterministically on every sample before it is substituted for $\hat{\mathfrak{R}}_N$; it is not used to define a sample-selected function class.

If the reference and budgets are chosen from an independent pilot sample, condition on that sample. The remaining N sequences retain their common distribution and independence, and the local class is fixed under this conditioning. Applying the theorem conditionally and then integrating the failure probability proves the unconditional claim. Deterministic floor certificates must hold for the conditioned class. Pilot-sample radius minima do not by themselves establish population floors.

For a countable family of predetermined references and constraints, apply the theorem with confidence $\delta\pi_a$ and union bound, as in Corollary F.3. This allows data selection among the listed classes and costs $\log(1/\pi_a)$ in the confidence logarithm. It does not justify an unrestricted same-sample reference chosen from a continuum. An absolute gap bound follows by applying the one-sided argument to both directions and using $\delta/2$ in each.

H DEPTH DEPENDENCE AND THE ROLE OF RADIUS GROWTH

The coefficient K combines two effects: downstream propagation, through the products Π_s , and the number of independently variable stages, through a sum of $2/3$ powers. Controlling the first effect does not automatically control the second.

Proposition H.1 (A uniform upper growth bound). *Every original or extended network in the stated parameter class satisfies*

$$\|H^s\|_{\infty, \circ} \leq B_0 + \sum_{u=1}^s \lambda_u \Lambda_u. \quad (46)$$

Proof. Every branch receives an input with maximum RMS radius at most one and has output radius at most Λ_u . Apply the triangle inequality to each residual update and iterate. $\square$

With fixed branch bounds and unscaled residuals, this rules out a geometrically growing *lower* radius floor at arbitrarily large depth. It does not prove a linear lower floor: branches may cancel, vanish, or point against the stream. A linear lower floor must be assumed or established by an additional argument.

Proposition H.2 (Propagation under a linear floor). *Suppose $\lambda_s = 1$, $\Lambda_s \leq \Lambda$, and $c_s \geq a + \zeta s$ for fixed $a, \zeta > 0$. Writing $p = \Lambda/\zeta$, for $0 \leq s \leq D$,*

$$\Pi_s \leq e^{\Lambda/a} \left(\frac{a + \zeta D}{a + \zeta s} \right)^p. \quad (47)$$

Thus residual propagation is polynomial in depth.

Proof. Use $\log(1+x) \leq x$ and $(c_{u-1}^2 + \nu)^{-1/2} \leq (a + \zeta(u-1))^{-1}$. The decreasing harmonic sum from $u = s+1$ to D is at most $a^{-1} + \zeta^{-1} \log((a + \zeta D)/(a + \zeta s))$. Exponentiate after multiplying by Λ . $\square$

Corollary H.3 (Accumulated loss-cover coefficient). *Under Proposition H.2, suppose d, r_{out} and all other non-depth constants are fixed, there are at most J_0 linear stages per sublayer, and $\kappa_j \leq u_0$ uniformly. Let*

$$K_{\text{B,loss}} = \left[\sum_j (\Pi_{\text{out}} \Pi_{s(j)} \kappa_j)^{2/3} \right]^{3/2}.$$

As $D \rightarrow \infty$, the following upper bounds hold:

$$K_{\text{B,loss}} = \begin{cases} O(D^{1/2}), & p < 3/2, \\ O(D^{1/2}(\log D)^{3/2}), & p = 3/2, \\ O(D^{p-1}), & p > 3/2. \end{cases} \quad (48)$$

Also $A = O(D^{p-1})$, and $K \leq \sqrt{2}(K_{\text{B,loss}} + \kappa_R)$.

Proof. The final normalizer gives $\Pi_{\text{out}} \leq 2r_{\text{out}}\sqrt{d}/(a + \zeta D)$. Combining this with (47) and the stage bounds yields

$$K_{\text{B,loss}} \leq C(a + \zeta D)^{p-1} \left[\sum_{s=1}^D (a + \zeta s)^{-2p/3} \right]^{3/2},$$

where C is independent of D . Integral comparison gives (48). The formula for A follows directly from (43); the last inequality follows from $(x^{2/3} + y^{2/3})^{3/2} \leq \sqrt{2}(x + y)$ for $x, y \geq 0$. $\square$

The exponents in Corollary H.3 describe coefficients in this bound, conditional on the entire list of fixed-budget and floor assumptions. They are not assertions about trained networks. For an arbitrarily deep nonempty family with these fixed upper growth bounds, (46) also requires $\zeta \leq \Lambda$. There is no guarantee that a given family satisfies any positive linear floor.

Proposition H.4 (A residual coefficient budget). *Suppose $c_s \geq c \geq 0$, $\Lambda_s \leq \Lambda$, and $\sum_s \lambda_s \leq A_0$. Then*

$$\Pi_0 \leq \exp\left(\frac{A_0 \Lambda}{\sqrt{c^2 + \nu}}\right).$$

If each sublayer has at most J_0 stages and $\ell_j R_j b_j \leq u_0$ uniformly, then, with all other constants fixed, $K_{\text{B,loss}} = O(A_0 \sqrt{D})$.

Proof. The propagation estimate follows by summing $\log L_s \leq \lambda_s \Lambda / \sqrt{c^2 + \nu}$. For the stage sum, $\kappa_j \leq \lambda_{s(j)} u_0$ and

$$\left(\sum_{s=1}^D \lambda_s^{2/3} \right)^{3/2} \leq \sqrt{D} \sum_{s=1}^D \lambda_s \leq A_0 \sqrt{D}$$

by Hölder's inequality. Bound all tail products and the final output factor by the constants just obtained. $\square$

The last proposition controls propagation independently of depth while leaving a possible $\sqrt{D}$ accumulation of capacity. A depth-free bound would need additional structure, for example shared parameters, a stronger summability constraint, or an argument exploiting cancellations that this product cover ignores. If the coefficients λ_s are learned rather than fixed, their admissible variation must also be covered or incorporated into the output-matrix constraint sets.

I COMPLETED EXPERIMENTAL PROTOCOLS AND SUPPLEMENTARY RESULTS

This appendix reports the experiments completed for this revision. The original E1–E4 baseline, the first architecture screen, and the three-seed extension are distinct data products. Shared conditions from the first screen are reused, not counted as additional independent runs.

Study	Seeds	Configurations	Directions	Sign draws
Exact construction (E1)	—	80 points	—	Exact sum
Dense baseline (E2–E4)	1	1	128	2,048
Initial architecture screen	1	10	32	1,024
Architecture extension	3 total	7	32	1,024

Table 3: Completed studies. Directions are paired with their negatives and include a reference model, giving family sizes 257 and 65. The extension includes the first screen’s seed for seven selected configurations.

I.1 DATA, TRAINING, ARCHITECTURE, AND PERTURBATIONS

For seed $b \in \{20260904, 20260905, 20260906\}$, separate CPU generators seeded by $b + 11, b + 23, b + 37$ generate pilot, fitting, and test data. Each generated row supplies 32 inputs and shifted next-token targets; the noisy modular recurrence is specified in the main text. There are 12,000 pilot, 512 fitting, and 2,000 test sequences. Sequence rows are the independent observations; correlated positions within a row are never counted as independent generalization samples. The three data groups are reused across architectures within each seed. The reference is trained on pilot sequences only. Fit losses are recorded each epoch, but training uses the fixed 12-epoch endpoint rather than selecting an epoch using these losses. No test data select the reference.

Training uses AdamW with learning rate 10^{-3} , weight decay 10^{-4} , batch size 128, and gradient clipping at norm 1. The model has no dropout or additive biases, uses fixed rotary maps, learned RMS gains, SwiGLU, untied embeddings/readout, and score scaling $1/\sqrt{d_h}$. For $d = 64, 128, 256$, head counts are 2, 4, 8 and $d_h = 32$. The feedforward width is $4d$. Logits are capped for the theorem’s loss by $10 \tanh(z/10)$; uncapped cross-entropy is retained separately in training histories. Normalization gains are learned during reference training and held fixed during the perturbation study.

The master seed controls initialization; the architecture trainer uses epoch-specific shuffling seeds $b + 101 + \text{epoch}$. The first baseline reuses its earlier completed checkpoint, whose loader uses one generator seeded at $b + 101$ throughout training. This implementation detail is retained in the code and provenance rather than silently retraining an already reported baseline. All 21 seed–architecture models have distinct fingerprints, and every archived training history has 12 epochs. Across these models, capped fit loss lies in $[0.5231, 0.5510]$, capped test loss in $[0.5338, 0.5549]$, fit accuracy in $[0.8845, 0.8901]$, and test accuracy in $[0.8835, 0.8879]$.

Gaussian directions are generated from $b + 1000$ and normalized with the notebook’s parameterwise relative-scale convention. The all-model group includes embeddings, attention, feedforward matrices, and readout; gains are held fixed. Readout changes have no effect on E2 hidden states but are included in E4 parameter budgets. The reference and both signs of each direction form the finite family. Direction banks are reused across scales. The empirical calculations use the achieved effective-matrix spectral and mixed-norm perturbations and total caps; η itself is not substituted for an achieved matrix norm. Every family is constructed independently of the evaluation signs. This samples a finite subset of the corresponding parameter neighborhood and does not optimize its continuous supremum.

E2 selects token index 16 (zero-based) from each input sequence. The architecture study uses a sign tensor seeded by $b + 44000$; its SHA-256 hash agrees across every architecture and scale within that seed and differs between seeds. Nested prefixes select the first m observations and signs for the baseline sample-size comparison. The dense baseline used a different sign stream at each scale, so its across-scale comparisons are not paired.

I.2 STATISTICAL SUMMARIES AND NUMERICAL AUDIT

For each fixed architecture, scale, sample size, and stage, the mean and sample standard deviation are computed across the three master seeds. Seed variation includes data, training, perturbation-family, and sign Monte Carlo variability; the present design does not identify separate variance components. Architecture contrasts are paired within seed. When reporting Π_0 or certificate slack, we take $\log_{10}$ within each seed before computing the mean and sample SD. This is not the logarithm of the arithmetic mean of the original quantities. Figure error bars are descriptive sample SDs, not confidence intervals for a population of seeds. In particular, three seeds cannot characterize rare training outcomes.

Within a fixed finite family, the empirical maximum over models is computed for each sign vector before averaging. The numerator and denominator of (23) are thus averages of their respective maxima, not an average of per-sign ratios. The notebook uses a paired bootstrap over sign draws (400 resamples in the architecture study; 1,000 in the dense baseline) to form pointwise 95% intervals for each ratio. These do not account for training or data variation and are not simultaneous 95% bands over all 612 rows. Their upper endpoints are reported as numerical evidence without claiming simultaneous coverage. The saved archives do not include all per-sign raw/normalized/frozen maxima, so the ordinary–frozen point differences are not assigned a paired-difference confidence interval.

The three-seed archive passes the ZIP integrity check; all individual condition CSVs agree with the consolidated CSVs. It contains exactly 60 condition rows and 612 stage rows, with no duplicate keys or numerical infinities. Its only missing condition fields are the two empty E4 error columns, which record no exceptions. Configuration fields agree with every row. The first seed’s seven-configuration outputs match the previous screen, including reference fingerprints and sign hashes. Ratios, logarithms, loss ranges, and test-minus-fit differences were recomputed from saved components. These checks establish consistency of the archived outputs; they do not replace independent model reruns or formal interval arithmetic.

I.3 E1: EXACT CONSTRUCTION

We evaluate $m \in \{8, 32, 128, 512, 2048\}$, $\nu \in \{0.1, 10^{-3}\}$, fixed $\rho \in \{0, 0.025, 0.1, (\sqrt{2} - 1)/2\}$, and shrinking schedules $\rho = a/\sqrt{m}$ and $\rho = a/m$ with $a \in \{0.25, 1\}$. This gives 80 deterministic values. For n_+ positive signs, put $S = 2n_+ - m$ and let k_+, k_- count coordinates with $a_i = t$ among positive and negative signs. With $q_0 = (1 + \nu)^{-1/2}$ and $q_1 = (1 + t^2 + \nu)^{-1/2}$, the squared normalized sum divided by d equals

$$V(n_+, k_+, k_-) = [q_0 S - (q_0 - q_1)(k_+ - k_-)]^2 + t^2 q_1^2 (k_+ + k_-). \quad (49)$$

It is convex on $[0, n_+] \times [0, m - n_+]$, so its maximum is attained at one of the four corners, all of which are admissible integer choices. The exact distortion is therefore

$$\frac{\sqrt{1 + \nu} \mathbb{E}_{n_+ \sim \text{Bin}(m, 1/2)} \sqrt{\max_{\text{four corners}} V(n_+, k_+, k_-)}}{\mathbb{E}_{n_+ \sim \text{Bin}(m, 1/2)} \sqrt{(2n_+ - m)^2 + t^2 m}}. \quad (50)$$

Binomial weights are computed stably. The notebook checks this reduction against exhaustive enumeration for $m \leq 5$. All archived exact values lie above the construction lower bound and below the theorem upper bound. An individual finite- m family ratio can be below one: the separate lower bound one in Theorem 3.2 is for the supremum over all admissible classes.

I.4 E2: SUPPLEMENTARY FINITE-FAMILY DIAGNOSTICS

In addition to the sampled floor $\hat{c}_s$, the code records

$$\begin{aligned} \hat{\rho}_s &= \frac{1}{2} \max_i \left[\max_{\theta} \|h_i(\theta)\|_{\circ} - \min_{\theta} \|h_i(\theta)\|_{\circ} \right], \\ \hat{\rho}_s^0 &= \max_{i, \theta} | \|h_i(\theta)\|_{\circ} - \|h_i(\theta^0)\|_{\circ} |. \end{aligned} \quad (51)$$

The first uses the minimal midpoint profile on the finite family; the second is the spread around the specified reference. Neither is a population floor or a supremum over the continuous parameter class. The frozen control is

$$h_i^{\text{fr}}(\theta) = h_i(\theta) / \sqrt{\|h_i(\theta^0)\|_{\circ}^2 + \nu}.$$

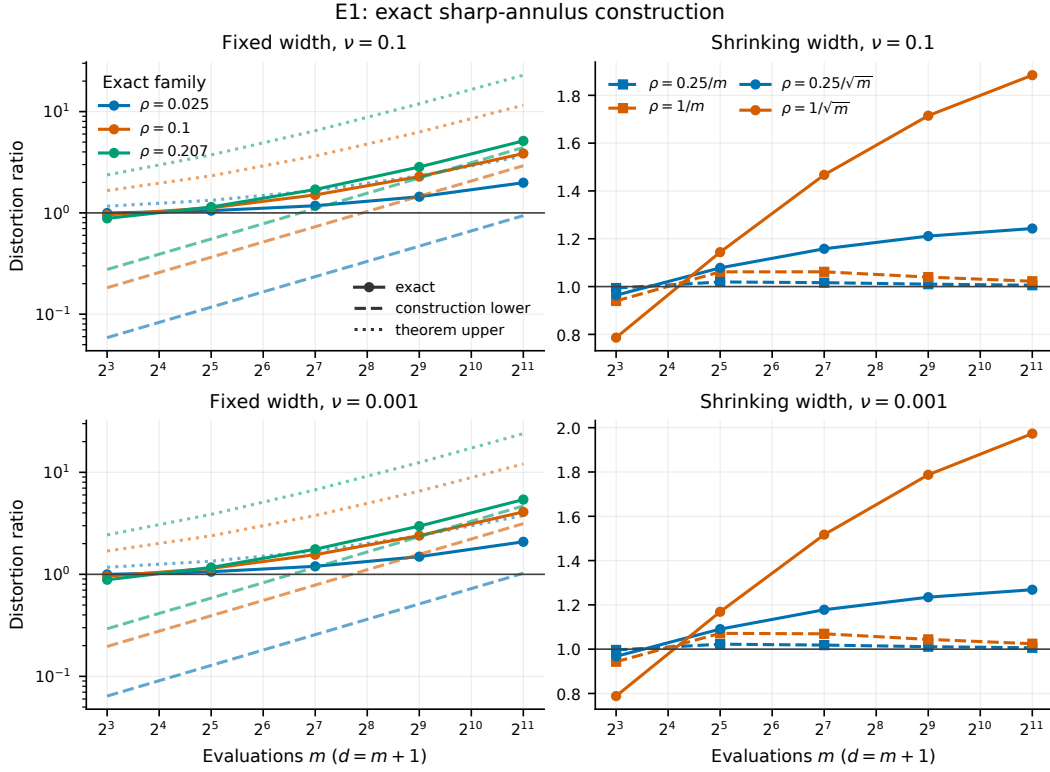


Figure 3: Exact E1 scaling. Fixed-annulus panels compare exact finite sums with construction lower and theorem upper bounds. Shrinking-annulus panels illustrate the threshold regimes. Every construction uses $d = m + 1$; this is not a fixed-hidden-width trained-model experiment.

ν	ρ	Exact family	Construction lower	Theorem upper
0.1	0.025	1.986	0.939	3.642
0.1	0.1	3.857	2.920	11.569
0.1	0.2071	5.136	4.420	22.890
0.001	0.025	2.084	1.025	3.770
0.001	0.1	4.091	3.140	12.080
0.001	0.2071	5.405	4.683	23.947

Table 4: E1 at $m = 2048$ and $d = m + 1$. The exact constructed-family ratio is below the theorem’s worst-case upper bound and above the analytic lower bound. The theorem’s separate lower bound of one concerns the supremum over classes, so an individual finite- m family ratio may be below one.

The raw, normalized, and frozen complexities are retained in the stage CSVs, along with their ratio estimates, radius summaries, and the profile-bound upper value. The vector process includes a nonzero reference singleton contribution, so small perturbations can leave the full-class ratio close to its reference value. The current archives do not isolate this contribution through a centered-increment experiment.

The earlier baseline used 128 directions plus negatives and reference (257 family members), five scales 10^{-4} , $3 \cdot 10^{-4}$, 10^{-3} , $3 \cdot 10^{-3}$, 10^{-2} , 2,048 sign draws, and seed 20260904. Its 45 stage-scale distortions lie between 0.449 and 0.954 and all pointwise upper endpoints are below one. The largest absolute ordinary-frozen point difference is $7.24 \cdot 10^{-4}$. The profile bound divided by observed normalized complexity ranges from 1.05 to 7.02. The three-seed study uses only the two larger scales and 32 directions; these outcomes are consistent, but are not a matched family-size sensitivity analysis because the Monte Carlo setup also differs.

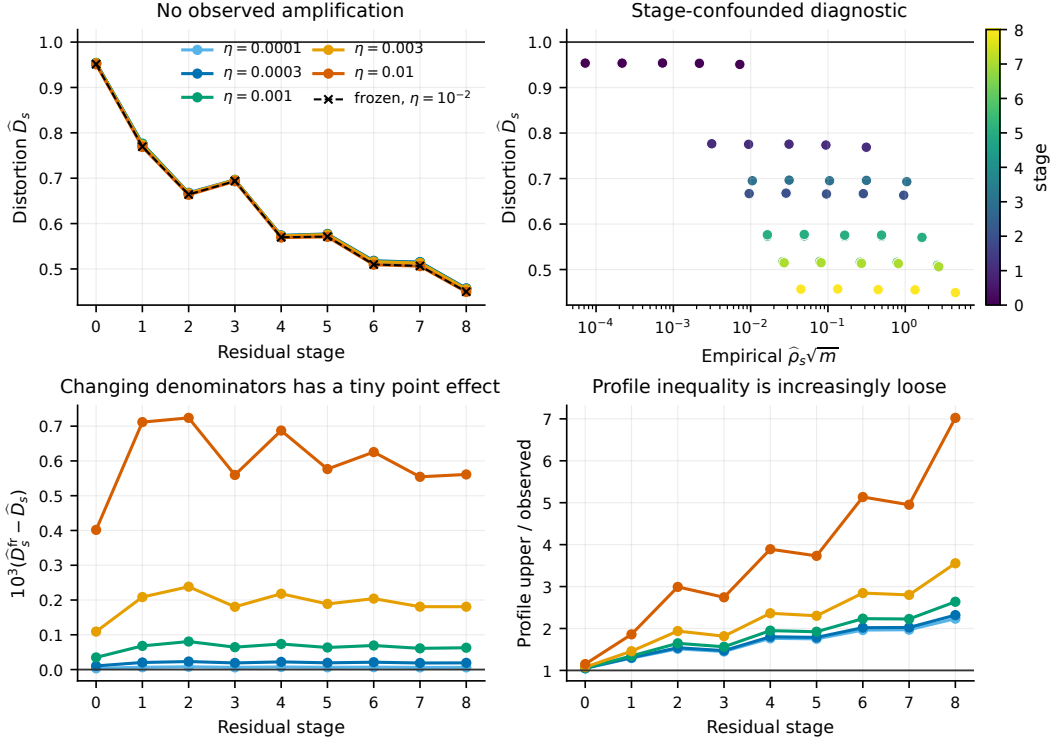


Figure 4: Earlier single-seed, 128-direction E2 baseline. Interval bars are conditional sign-bootstrap uncertainty. Stage colors confound the distortion–radius plot, so it is not evidence of a causal relation. This study is separate from the three-seed architecture averages.

Configuration	$\max_s \hat{D}_s$	$\log_{10} \Pi_0$	$\log_{10} \text{slack}$	Test loss
Baseline	0.922 ± 0.033	41.43 ± 0.13	38.75 ± 0.17	0.5428 ± 0.0081
Depth 2	0.916 ± 0.020	20.80 ± 0.11	18.35 ± 0.04	0.5392 ± 0.0058
Depth 8	0.911 ± 0.022	82.99 ± 0.02	80.37 ± 0.14	0.5461 ± 0.0060
Width 64	0.893 ± 0.019	39.01 ± 0.17	36.53 ± 0.17	0.5383 ± 0.0056
Width 256	0.927 ± 0.015	45.09 ± 0.11	42.07 ± 0.26	0.5451 ± 0.0059
Scaled L=8	0.920 ± 0.011	73.39 ± 0.02	71.38 ± 0.03	0.5477 ± 0.0066
$\nu = 10^{-3}$	0.932 ± 0.029	33.46 ± 0.14	31.09 ± 0.18	0.5420 ± 0.0074

Table 5: Three-seed means $\pm$ sample SD at $m = 512, \eta = 10^{-2}$. The maximum is taken over stages within each seed before averaging. Propagation and slack are summarized on the base-10 log scale.

I.5 E3: FLOORS, DISPLACEMENT, AND THE EARLIER STAGewise AUDIT

The zero-floor recursion is evaluated using achieved perturbation budgets and ambient effective-matrix caps. Positive candidates are fractions 0.25, 0.5, 0.75 of the minimum reference radius over the fit set and all positions. Each is tested using the sufficient inequalities of Theorem 4.2. Every tested positive candidate fails in all 42 distinct architecture–seed–scale settings. The additional baseline m prefixes alter E2 and E4, but E3 still uses the full 512-sequence fit set, so its entries across these prefixes must not be counted as independent floor tests. An infeasible candidate is reported; it is not substituted into E4 as a certified floor.

The earlier dense baseline includes the complete stagewise certificate table at $\eta = 10^{-2}$ (Figure 5). Measured final displacement is 0.4169, whereas the zero-floor final certificate is $3.70 \cdot 10^{38}$, for slack $8.89 \cdot 10^{38}$. The largest certified branch Lipschitz constant is 1474.9, zero-floor normalization

Configuration	Seed	$\max_s \hat{D}_s$	$\log_{10} \Pi_0$	$\log_{10} \text{slack}$	$U_N/(B/2)$
Baseline	20260904	0.9522	41.58	38.95	340.3
Baseline	20260905	0.9260	41.39	38.67	339.5
Baseline	20260906	0.8866	41.33	38.62	339.2
Depth 2	20260904	0.9391	20.93	18.40	243.9
Depth 2	20260905	0.9089	20.74	18.34	242.8
Depth 2	20260906	0.9015	20.72	18.32	242.7
Depth 8	20260904	0.9191	82.97	80.44	477.3
Depth 8	20260905	0.9283	83.02	80.21	477.5
Depth 8	20260906	0.8867	83.00	80.45	477.5
Width 64	20260904	0.8719	38.83	36.33	235.3
Width 64	20260905	0.8988	39.02	36.63	235.7
Width 64	20260906	0.9084	39.17	36.62	236.3
Width 256	20260904	0.9338	45.03	42.16	496.9
Width 256	20260905	0.9386	45.22	42.27	497.7
Width 256	20260906	0.9099	45.03	41.77	496.7
Scaled L=8	20260904	0.9220	73.38	71.38	449.4
Scaled L=8	20260905	0.9294	73.39	71.34	449.5
Scaled L=8	20260906	0.9081	73.41	71.41	449.6
$\nu = 10^{-3}$	20260904	0.9580	33.61	31.27	302.3
$\nu = 10^{-3}$	20260905	0.9377	33.41	31.10	301.4
$\nu = 10^{-3}$	20260906	0.9014	33.35	30.90	301.0

Table 6: Individual seed values at $m = 512, \eta = 10^{-2}$. These rows permit direct inspection of the averages and the changes in architecture ordering. They are not additional observations beyond Table 5.

s	Stream	Observed max.	$\Delta_s (c_s = 0)$	Slack	$\Delta_s (c_s = .75a_s)$	Slack
0	Embedding	9.45×10^{-4}	9.45×10^{-4}	1	9.45×10^{-4}	1
1	Block 1 attention	0.0583	277	4.76×10^3	22.2	380
2	Block 1 feedforward	0.128	1.47×10^7	1.15×10^8	5.41×10^3	4.23×10^4
4	Block 2 feedforward	0.214	3.46×10^{17}	1.62×10^{18}	9.13×10^8	4.27×10^9
6	Block 3 feedforward	0.255	1.05×10^{28}	4.13×10^{28}	1.15×10^{14}	4.5×10^{14}
8	Block 4 feedforward	0.417	3.7×10^{38}	8.89×10^{38}	1.11×10^{19}	2.67×10^{19}

Table 7: E3 at $\eta = 10^{-2}$. The zero-floor recursion remains a valid displacement certificate even when it cannot derive a positive floor. The $0.75a_s$ candidate fails feasibility after the embedding stage. The stage-zero ratio differs from one by 4.3×10^{-7} because the measured forward pass and norm budget use different floating-point accumulation; no substantive violation occurs at a relative tolerance of 10^{-6} .

contributes $1/\sqrt{\nu} = 316.2$, and the largest residual multiplier is $4.66 \cdot 10^5$. The eight-sublayer product is $3.79 \cdot 10^{41}$. These constants describe a uniform parameter class, whereas the observed maximum is over the sampled finite family. Their difference therefore combines conservativeness of the argument with incomplete search of that class; it does not measure the exact tightness of the optimal uniform bound.

At stage zero, the computed certificate undershoots the measured displacement by relative $4.29 \cdot 10^{-7}$. The discrepancy is within the stated 10^{-6} floating-point tolerance and is retained in the audit rather than silently set to zero. Later zero-floor stages exceed the observations by many orders of magnitude. This is numerical validation, not certified interval arithmetic. Candidate-floor flags do not determine whether the unconditional zero-floor displacement statement holds.

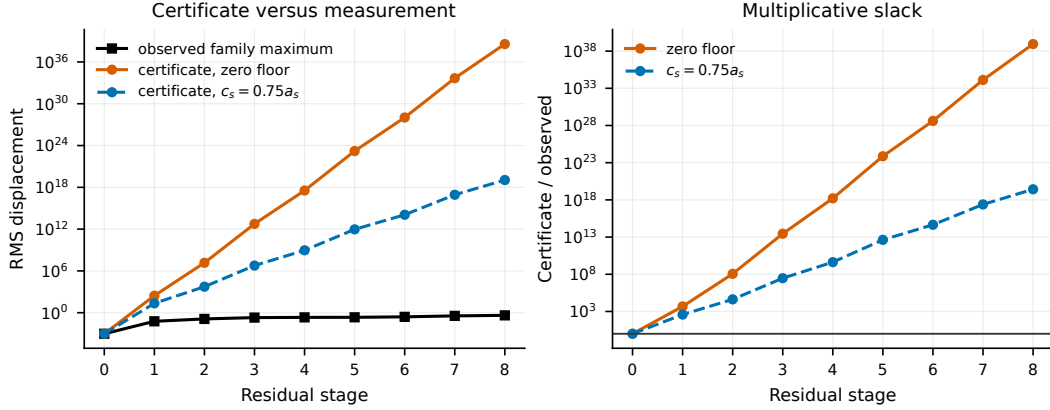


Figure 5: Earlier 128-direction baseline at $\eta = 10^{-2}$. Positive candidate floors reduce the formal recursion but fail the sufficient feasibility test; only the unconditional zero-floor calculation can be used without an independently established floor.

The initial ten-configuration screen also included scaled $L = 2$, scaled $L = 4$, and $\nu = 10^{-4}$, which were not repeated over the two new seeds. At $\eta = 10^{-2}$ their log-products were respectively 19.57, 37.90, and 37.58. None produced a positive-floor certificate or a nonvacuous E4 bound. The smallest slack in that wider, single-seed screen was $2.62 \cdot 10^{17}$ for scaled $L = 2$ at $\eta = 10^{-3}$. The minimum $1.87 \cdot 10^{18}$ in the three-seed main text refers to the seven-configuration subset, so the two minima concern different grids. Full first-screen rows are included in the companion data.

I.6 E4: BOUND DECOMPOSITION AND SAMPLE-SIZE CURVES

All E4 comparisons use the same decoder, capped loss, confidence parameter $\delta = 0.05$, and sample unit. Local perturbation budgets and their matched ambient/global budgets use the same propagation constants and zero floors. The calculation includes embeddings, all linear branch stages, readout, entropy logarithms, and the confidence term. It is an internal comparison of the present formulas, not a ranking of bounds for incompatible architectures from different papers.

The numerical optimizer searches $\log(\alpha/B) \in [-40, 0]$ and explicitly checks the endpoint $\alpha = B$. We report its returned raw value U_N , the intermediate $\Phi_N = \min\{B/2, U_N\}$, and the final $\min\{B, 2\Phi_N + 3B\sqrt{\log(2/\delta)/(2N)}\}$. The field named `untruncated_gap_bound` in the earlier baseline is after the Φ_N cap but before the final B cap; the architecture code calls it `pre_final_cap_gap`. At $N = 512$ this intermediate quantity is 25.8975, and the reported value is $B = 21.94591$. The entire observed raw range is well above $B/2$. A derivative test also rules out a missed interior optimum for the archived coefficient values.

Lemma I.1 (Endpoint regime of the entropy objective). *Let $F_N(\alpha)$ denote the expression inside the infimum in (21), with $B, N, \mathcal{L} > 0$ and $A_{\text{loc}}, P, K_{\text{loc}} \geq 0$. If*

$$K_{\text{loc}} \geq \frac{B\sqrt{N}}{96\sqrt{\mathcal{L}}},$$

then F_N is nonincreasing on $(0, B]$ and $\inf_{0 < \alpha \leq B} F_N(\alpha) = F_N(B) \geq 4B$. Consequently $\Phi_N = B/2$ and the reported gap bound is B .

Proof. The embedding term is nonincreasing in α . Differentiating the other two terms gives, for $0 < \alpha < B$,

$$F'_N(\alpha) \leq 4 - \frac{384K_{\text{loc}}\sqrt{\mathcal{L}}}{\sqrt{N}\alpha} \leq 4 - \frac{384K_{\text{loc}}\sqrt{\mathcal{L}}}{\sqrt{N}B} \leq 0.$$

Continuity at B and nonnegativity of the remaining endpoint term complete the argument. $\square$

For all 60 archived conditions, the ratio $96K_{\text{loc}}\sqrt{\mathcal{L}}/(B\sqrt{N})$ exceeds $8.00 \cdot 10^{20}$. Thus the coefficient values place the evaluated formula firmly in this endpoint regime. This establishes the optimum for those coefficient values, while the underlying norm measurements remain floating-point calculations.

In the dense baseline, local $K = 7.53 \cdot 10^{42}$ is 86.5 times smaller than matched global K , while raw U_N changes from 3820.46 to 3733.70. The minimizing candidate is $\alpha = B$. At this endpoint the stage-integral term vanishes, but the remaining embedding-entropy term still depends on the propagation product through A_{loc} . The first-block query/key terms account for 93.8% of the $\sum_j w_j^{2/3}$ allocation sum. This percentage describes the covering coefficient, not the share of the optimized scalar objective; the distinction matters at an endpoint optimum.

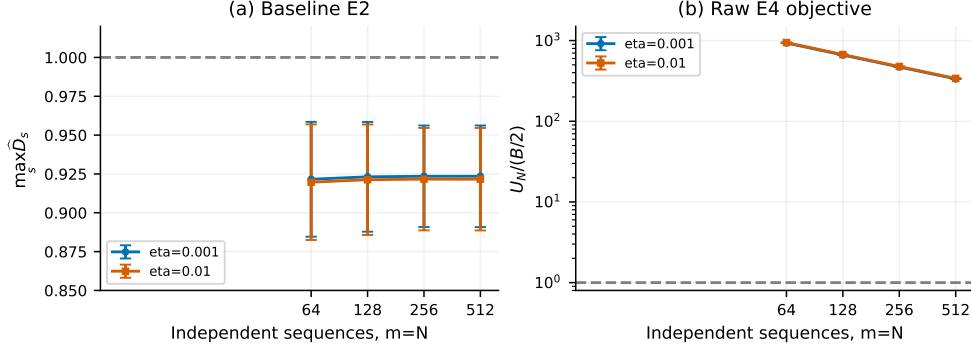


Figure 6: Baseline sample-size curves, mean $\pm$ sample SD across three seeds. E2 maxima change little across nested prefixes. The raw E4 objective decreases with N but remains hundreds of times above $B/2$. Sample-size points within a seed are dependent.

η	$m = N$	$\max_s \hat{D}_s$	$U_N/(B/2)$
$1e-03$	64	0.922 ± 0.037	933.3 ± 1.6
$1e-03$	128	0.923 ± 0.035	662.3 ± 1.1
$1e-03$	256	0.924 ± 0.033	470.7 ± 0.8
$1e-03$	512	0.923 ± 0.033	335.1 ± 0.6
$1e-02$	64	0.920 ± 0.037	946.0 ± 1.6
$1e-02$	128	0.921 ± 0.036	671.3 ± 1.1
$1e-02$	256	0.922 ± 0.033	477.0 ± 0.8
$1e-02$	512	0.922 ± 0.033	339.6 ± 0.5

Table 8: Values underlying Figure 6. Each entry reports the across-seed mean $\pm$ sample SD; no E3 retest is implied.

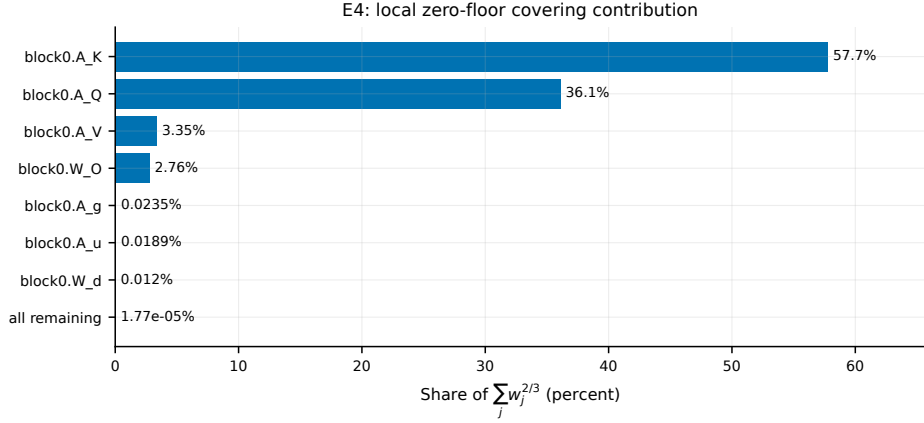


Figure 7: Earlier dense baseline: contributions to the local zero-floor $2/3$ -power allocation sum. These are not contributions to U_N evaluated at its minimizing candidate.

Component	s	Budget	w_j	Share of $\sum w^{2/3}$
block0.A_K	1	1.55	3.31×10^{42}	57.7435%
block0.A_Q	1	1.55	1.63×10^{42}	36.0916%
block0.A_V	1	1.34	4.61×10^{40}	3.3460%
block0.W_O	1	1.36	3.46×10^{40}	2.7644%
block0.A_g	2	5.23	2.71×10^{37}	0.0235%
block0.A_u	2	5.18	1.96×10^{37}	0.0189%
block0.W_d	2	1.52	9.88×10^{36}	0.0120%

Table 9: Largest E4 local zero-floor components. The first attention query/key terms contribute about 93.8% of the optimized $2/3$ -power sum, pinpointing the principal source of the covering constant.

I.7 REMAINING EXPERIMENTS AND LIMITS OF REPRODUCIBILITY

Further experiments should target five remaining uncertainties.

1. *Adversarial finite families.* Optimize directions on search signs, freeze the family, and evaluate on fresh signs and sequences. Report the objective, restarts, achieved caps, and random-family controls. Failure to find amplification is not a proof that it is impossible.
2. *Family-size and reference controls.* Use fixed checkpoints, nested banks of 32 and 128 streamed directions, and common signs. Compare initialization with trained checkpoints; report singleton and centered-increment diagnostics separately from the theorem’s process.
3. *Full-support radius validation.* Enumerate all 256 input sequences of a $V = 4, T = 4$ task. This supplies exhaustive numerical reference minima; a formally certified floor also requires conservative numerical error bounds.
4. *Text data.* Use WikiText-2 raw text (Merity et al., 2016) with document-disjoint splits, fixed byte preprocessing, and dataset hashes. Report bits per byte and capped loss; within-document windows are not independent theorem samples. This experiment remains unrun.
5. *Mechanism and bound ablations.* Compare sampled Jacobians with the uniform product; fix embeddings or readout separately. Derivative diagnostics require a uniform proof. Cross-paper comparisons must share architectural and statistical assumptions.

The package reproduces summaries from archived measurements using the included analysis script. Architecture archives omit checkpoints, direction banks, hidden-state caches, and per-sign maxima needed for forward-pass reconstruction and paired-difference analysis. The supplied notebook code is included; models were not rerun for this revision.